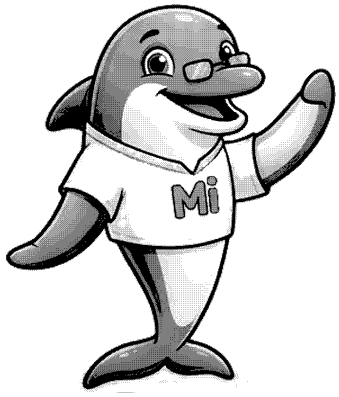
Math Inquirers

# Extension Workbook

## with answers

## Level 4

Self-study | Ages 9-11




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 4**
Self-study practice with answers.


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## Good to know

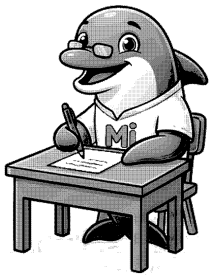
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# Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

## How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

## For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 25 to 30 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 79: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- You will need: a pencil, a rubber, a ruler and a 180 degree protractor (Unit 6).
- Beyond Level 4 is ahead of school. You may skip it.
- The angles in Unit 6 are printed at real size.

## The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

# Contents


| | | |
|---|---|---|
| | **About this book**<br>Copyright and licence | page 2 |
| | **Welcome** | page 3 |
| | **My pace plan** | page 5 |
| 1 | **Factor Hunters**<br>Factor hunters · follows Holiday Program Lesson 1 · answers on page 80 | page 6 |
| 2 | **Big Number Puzzles**<br>Big number puzzles · follows Holiday Program Lesson 2 · answers on page 82 | page 13 |
| 3 | **Operation Detectives**<br>Operation detectives · follows Holiday Program Lesson 3 · answers on page 84 | page 20 |
| 4 | **Fraction Frontier**<br>Fraction frontier · follows Holiday Program Lesson 4 · answers on page 86 | page 27 |
| 5 | **Decimal Detectives**<br>Decimal detectives · follows Holiday Program Lesson 5 · answers on page 88 | page 34 |
| 6 | **Angle Turns**<br>Angle turns · follows Holiday Program Lesson 6 · answers on page 91 | page 41 |
| 7 | **Area Architects II**<br>Area architects II · follows Holiday Program Lesson 7 · answers on page 93 | page 48 |
| 8 | **Symmetry and Nets**<br>Symmetry and nets · follows Holiday Program Lesson 8 · answers on page 95 | page 55 |
| 9 | **Data Detectives**<br>Data detectives · follows Holiday Program Lesson 9 · answers on page 97 | page 62 |
| 10 | **Puzzle Day**<br>Puzzle day · follows Holiday Program Lesson 10 · answers on page 99 | page 69 |
| | **My progress** | page 76 |
| | **My certificate** | page 77 |
| | **Answers**<br>Check your work, with short reasons | page 79 |

# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 25 to 30 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | Factor Hunters | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 2 | Big Number Puzzles | 13-14 | 15-16 | 17 and 19 | 18 | ☐ |
| 3 | Operation Detectives | 20-21 | 22-23 | 24 and 26 | 25 | ☐ |
| 4 | Fraction Frontier | 27-28 | 29-30 | 31 and 33 | 32 | ☐ |
| 5 | Decimal Detectives | 34-35 | 36-37 | 38 and 40 | 39 | ☐ |
| 6 | Angle Turns | 41-42 | 43-44 | 45 and 47 | 46 | ☐ |
| 7 | Area Architects II | 48-49 | 50-51 | 52 and 54 | 53 | ☐ |
| 8 | Symmetry and Nets | 55-56 | 57-58 | 59 and 61 | 60 | ☐ |
| 9 | Data Detectives | 62-63 | 64-65 | 66 and 68 | 67 | ☐ |
| 10 | Puzzle Day | 69-70 | 71-72 | 73 and 75 | 74 | ☐ |
| 11 | Catch-up week | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

### Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

# Learn it: Factors and multiples

**My goal. I can list every factor of a number in order. I can find common factors and common multiples and use them to solve sharing and timetable puzzles.**

A factor goes into a number exactly. A multiple is in a times table. Write factors in pairs, in order, and stop when the pairs meet.

## Worked example 1

**Find all the factors of 54.**

| Try | Pair |
|---|---|
| 1 | 1 × 54 |
| 2 | 2 × 27 |
| 3 | 3 × 18 |
| 4, 5 | no |
| 6 | 6 × 9 |
| 7, 8 | no |

1. Try 1, 2, 3 in order. Write a pair each time the number goes in.
2. 4 and 5 do not go into 54. 6 × 9 = 54 is a pair.
3. 7 and 8 do not go in, and the next number, 9, is already used. The pairs have met, so stop.

**Answer: The factors of 54 are 1, 2, 3, 6, 9, 18, 27, 54.**

## Worked example 2

**Lamp A flashes every 4 seconds. Lamp B flashes every 10 seconds. They flash together now. After how many seconds do they next flash together?**

| Lamp | Flashes after these seconds |
|---|---|
| A | 4, 8, 12, 16, 20 |
| B | 10, 20 |

1. List the multiples of each number.
2. Find the first number in both lists.

**Answer: They next flash together after 20 seconds.**

## How to hunt

1. Factors: try 1, 2, 3 in order. Stop when the pairs meet.
2. Common factors: list both numbers, then pick the numbers in both lists.
3. Common multiples: list the multiples of both. The first number in both lists is the least.

## Good to know

- A pair like 6 × 6 counts as one factor.
- So 36 has 9 factors, an odd number.

## Your turn

Answer at the back of the book.

**Try** **Write all the factors of 45, in order.**

_____

Name _____

## Start here

Do these first. Take your time.

**1** **Fill in the factor pairs of 56. Some numbers do not go in.**

**(a)** 1 × 56  2 × ☐  4 × ☐  7 × ☐

**(b)** How many factors does 56 have? ☐ factors

**2** **Write all the factors of 66, in order from the smallest.**

_____

**3** **A class has 32 children. The teacher makes equal teams. Every child is in a team. There is more than 1 team and more than 1 child in a team. List every team size that is possible.**

_____

Working: _____

**4** **Circle True or False.**

**(a)** 7 is a factor of 91.  True  False

**(b)** 54 is a multiple of 8.  True  False

**(c)** 9 is a factor of 108.  True  False

**(d)** 65 is a multiple of 13.  True  False

**5** **Write the first 6 multiples of each number. Then find the least number that is in both lists.**

**(a)** Multiples of 9: _____

**(b)** Multiples of 12: _____

**(c)** Least common multiple: ☐

## Start here

Keep going. Check your work.

**6** **Find the common factors of 28 and 42.**

**(a)** Factors of 28: _____

**(b)** Factors of 42: _____

**(c)** Common factors: _____

**(d)** Greatest common factor: ☐

**7** **Ravi has 30 red marbles and 45 blue marbles. He fills equal bags and no marble is left out. Each bag holds the same number of red marbles, and the same number of blue marbles.**

**(a)** What is the most bags he can fill? ☐ bags

**(b)** Then each bag has ☐ red and ☐ blue.

Factors of 30: _____

Factors of 45: _____

**8** **Kumar jogs every 10 days. Aisha jogs every 15 days. They both jog today.**

**(a)** They next jog together after ☐ days.

**(b)** How many more times do they jog together in the next 100 days? ☐ times

Kumar: 10, 20, _____

Aisha: 15, 30, _____

**9** **Who am I? I am between 50 and 100. Both 6 and 7 go into me exactly.**

**I am** ☐ **.**

## Start here

Nearly there. You can do it.

**10** **Count the factors of each number.**

| Number | 8 | 14 | 18 | 27 | 32 | 49 |
|---|---|---|---|---|---|---|
| How many factors? | | | | | | |

**(a)** Write the number of factors under each number.

**(b)** Which number in the table has exactly 3 factors? ☐

**11** **Mei buys pens in packs of 6 and rubbers in packs of 8. She wants the same number of pens and rubbers.**

**(a)** What is the fewest of each she can buy? ☐ of each

**(b)** How many packs of pens is that? ☐ packs

**(c)** How many packs of rubbers is that? ☐ packs

Multiples of 6: _____

Multiples of 8: _____

**12** **I am a factor of 60. I am a multiple of 5. I am greater than 10 and less than 30. Write every number that fits.**

_____ _____

**13** **Find out which numbers have an odd count of factors.**

**(a)** How many factors does 64 have? ☐

**(b)** How many factors does 81 have? ☐

**(c)** Which has an odd count of factors, 72 or 49? 72 49

## Think harder

Hints are upside down. Try first!

**1** **A number between 10 and 100 is in the 3 times table and in the 8 times table. Its tens digit is greater than its ones digit. Its digits add up to 9. What is the number?**

☐

Working: _____

_____

**Hint.** Which numbers are in both the 3 and 8 times tables?

**2** **Ben has more than 50 stickers and fewer than 100. He packs them in 4s and 3 are left over. He packs them in 5s and 3 are left over. He packs them in 6s and 3 are left over. How many stickers does he have?**

☐

Working: _____

_____

_____

**Hint.** Take 3 away. What number is in the 4, 5 and 6 times tables?

**3** **Two gears have marks that touch at the start. Gear A has 12 teeth. Gear B has 18 teeth. They turn together until the marks touch again.**

**(a)** How many teeth go past the touching point? ☐ teeth

**(b)** Gear A turns ☐ times and gear B turns ☐ times.

Working: _____

_____

_____

**Hint.** The number of teeth is a multiple of both 12 and 18.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **What is the least number that has exactly 10 factors? Show the factors you found.**

☐

My factor lists: _____

**Hint.** Count the factors of some numbers between 20 and 60.

★ **Beyond Level 4: Three common multiples**

**2** **Three lighthouses flash at 12:00:00. Lighthouse A flashes every 6 seconds, B every 8 seconds and C every 9 seconds.**

**(a)** They next flash together after ☐ seconds.

**(b)** How many times do they flash together in the next 10 minutes? Do not count 12:00:00. ☐ times

**Hint.** A common multiple of 3 numbers is in all 3 lists.

**3** **Muffins are sold only in boxes of 4 and boxes of 7. Some numbers of muffins cannot be bought exactly. For example, you cannot buy exactly 5. What is the greatest number that cannot be bought exactly?**

☐

Boxes I can make: _____

**Hint.** Write the numbers 1 to 30 and cross out those you can make.

**4** **A rectangle has an area of 84 cm². Its length and width are whole numbers of centimetres. Its perimeter is 38 cm. Find its length and its width.**

**The length is the greater side. Length ☐ cm Width ☐ cm**

**Hint.** Write the factor pairs of 84. Find the pair that adds to 19.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Write all the factors of 52, in order.**

_____

Working: _____

**2** **Siti swims every 6 days. Wei swims every 9 days. They both swim today. In how many days will they next swim on the same day?** ☐ **days**

Working: _____

**3** **Kumar has 40 red pens and 64 blue pens. He makes identical sets with none left over. How many sets can he make at most?** ☐ **sets**

Working: _____

**4** **I am a multiple of 7. I am a factor of 84. My digits add up to 5. Who am I?**

☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

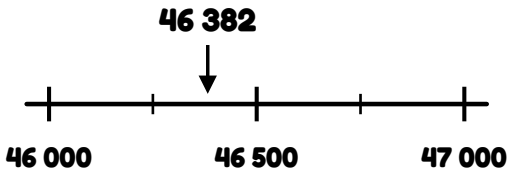
# Learn it: Numbers to 100 000

**My goal.** I can round numbers to the nearest 10, 100 and 1000. I can work backwards from a rounded number and solve digit puzzles.

To round a number, find the two neighbours it lies between. Halfway rounds up. To work backwards from a rounded number, find the smallest and the largest numbers that round to it.

## Worked example 1

**Round 46 382 to the nearest 1000.**

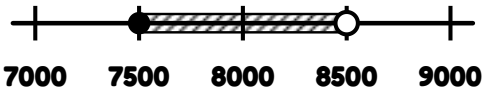



1. 46 382 is between 46 000 and 47 000.
2. The halfway mark is 46 500. 46 382 is before it.

**Answer: 46 382 rounds to 46 000.**

## Worked example 2

**A number rounds to 8000 to the nearest 1000. What are the smallest and the largest it can be?**




1. 7500 is halfway, so it rounds up to 8000.
2. 8500 rounds up to 9000, so the last number is 8499.

**Answer: Smallest 7500 and largest 8499.**

## How to round

1. Find the 2 neighbours, such as 46 000 and 47 000.
2. Find the halfway number. Halfway rounds up.
3. To work backwards, find where the rounding changes.

## Place value

| 10 000s | 1000s | 100s | 10s | 1s |
|---|---|---|---|---|
| 3 | 5 | 2 | 0 | 6 |

- 35 206 = 30 000 + 5000 + 200 + 6
- Halfway rounds up: 46 500 goes up to 47 000.

## Your turn

Answer at the back of the book.

Try **Round 52 718 to the nearest 100.** ☐

Name _____

## Start here

Do these first. Take your time.

**1** Round each number. Write the answers in the table.

| Number | Nearest 10 | Nearest 100 | Nearest 1000 |
|---|---|---|---|
| 38 746 | | | |
| 70 049 | | | |
| 9996 | | | |

**2** Write the value of the digit 6 in each number.

(a) 46 281: ☐ (b) 60 912: ☐ (c) 83 062: ☐

**3** Write each number in figures.

| In words | In figures |
|---|---|
| forty-two thousand and seven | |
| sixty thousand, nine hundred | |
| eight thousand and thirty | |

**4** Write the numbers in order. Begin with the greatest.

58 304 58 340 58 034 58 430

_____

## Start here

Keep going. Check your work.

**5** **The arrows point to numbers on the line. Each small step is 100.**




**(a)** A is ☐

**(b)** B is ☐

**(c)** Which thousand is B closer to? ☐

**6** **A match had 36 482 people watching. Round the number of people to the nearest 1000 and to the nearest 100.**

**(a)** Nearest 1000: ☐ **(b)** Nearest 100: ☐

**7** **A town has 12 000 people after rounding to the nearest 1000. Find the least number of people it could have, and the greatest.**

**(a)** Least: ☐ **(b)** Greatest: ☐

Working: _____

**8** **A number rounds to 4700 to the nearest 100.**

**(a)** The smallest it can be is ☐ **(b)** The largest it can be is ☐

**(c)** How many whole numbers could it be? ☐

## Start here

Nearly there. You can do it.

**9** **A ferry carries 4018 people on each trip. It makes 5 trips.**

**(a)** Round 4018 to the nearest 1000. Estimate the total: ☐

**(b)** Find the exact total: ☐

**(c)** The exact total is more less than the estimate.

Working: _____

**10** **I have 5 digits. Round me to the nearest 1000 and you get 52 000. My hundreds digit is 7. My tens digit is 5 and my ones digit is 2. Who am I?**

**I am** ☐ .

**11** **Use the digit cards 2, 4, 0, 7 and 9. Use each card once.**

**(a)** The greatest number: ☐

**(b)** The least number: ☐

**(c)** The greatest odd number: ☐

**12** **Find the rule. Write the next 2 numbers.**

**(a)** 31 600, 32 600, 33 600, ☐ , ☐

**(b)** 70 400, 68 900, 67 400, ☐ , ☐

**(c)** 4997, 5000, 5003, 5006, ☐ , ☐

**13** **Circle the correct sign.**

**(a)** 45 009 > < 45 090

**(b)** 8999 > < 10 000

**(c)** 62 100 > < 62 010

**(d)** 30 000 + 700 > < 30 070

## Think harder

Hints are upside down. Try first!

**1** **Cards: 1, 3, 4, 6, 8. Use each card once. Make the 5-digit number nearest to 40 000.**

Working: _____

**Hint.** Check one number just below 40 000 and one just above.

**2** **Two crowds were counted. To the nearest 1000, one crowd was 41 000 and the other was 28 000.**

**(a)** **The smallest possible total of both crowds:** ☐

**(b)** **The largest possible total of both crowds:** ☐

Working: _____

**Hint.** Add the 2 smallest numbers. Then add the 2 largest.

**3** **A 5-digit number has 5 different digits. It rounds to 70 000 to the nearest 10 000.**

**(a)** **The greatest it can be:** ☐

**(b)** **The least it can be:** ☐

Working: _____

**Hint.** Numbers that round to 70 000 run from 65 000 to 74 999.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Ravi rounds 36 449 in 3 steps. First to the nearest 10, then that answer to the nearest 100, then that answer to the nearest 1000.**

**(a)** After the 3 steps he gets ☐

**(b)** Round 36 449 to the nearest 1000 in one go: ☐

**(c)** Rounding 1 step at a time gives a wrong answer for some 4-digit numbers. What is the least 4-digit number where it does? ☐

**Hint.** Round a few numbers just below 1500 in steps.

★ **Beyond Level 4: Counting by place**

**2** **Ben writes the whole numbers from 1 to 1000. How many of them contain at least one 7?**

☐

Working: _____

**Hint.** Count the numbers with no 7. Choose each digit in turn.

**3** **Two patterns are woven together: 12 000, 3, 11 000, 6, 10 000, 9, and so on.**

**(a)** The 9th number is ☐

**(b)** The 10th number is ☐

**Hint.** Take every second number. You get 2 patterns.

**4** **Some numbers round to 7000 to the nearest 1000. Siti says they all round to 7000 to the nearest 100 too.**

**(a)** Is she right? Yes No

**(b)** How many of the whole numbers that round to 7000 to the nearest 1000 also round to 7000 to the nearest 100? ☐

**Hint.** Find the first and last numbers for each rounding.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Round 25 962 to the nearest 10 ☐, 100 ☐, 1000 ☐**

Working: _____

**2** **To the nearest 100, 8300 people came. Least: ☐ Greatest: ☐**

Working: _____

**3** **Cards: 4, 0, 7, 1, 8. Make the least odd 5-digit number. ☐**

Working: _____

**4** **Next 2 numbers: 53 200, 51 700, 50 200, ☐, ☐**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

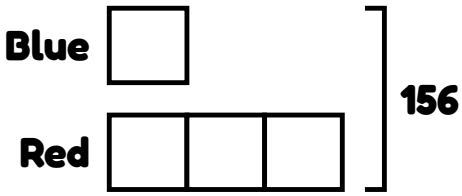
# Learn it: Four operations

**My goal.** I can solve multi-step problems with the four operations. I can use bar models for "times as many" and work backwards through a number machine.

A bar model shows a "times as many" story. Each equal part is one unit. To go back to the start of a number machine, undo each step in the opposite order.

## Worked example 1

**Mr Tan has 3 times as many red pens as blue pens. He has 156 pens in all. How many blue pens does he have?**

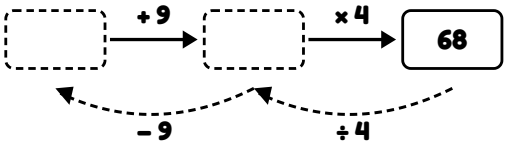



1. Blue is 1 unit and red is 3 units. That is 4 units.
2. 4 units = 156, so 1 unit = 156 ÷ 4 = 39.

**Answer: He has 39 blue pens.**

## Worked example 2

**I think of a number. I add 9 and then multiply by 4. The answer is 68. What is my number?**




1. Undo the steps in the opposite order.
2. Undo × 4 with ÷ 4: 68 ÷ 4 = 17. Undo + 9 with – 9: 17 – 9 = 8.

**Answer: My number is 8. Check: 8 + 9 = 17 and 17 × 4 = 68.**

## How to solve it

1. Bar models: draw one bar for each quantity. Count the units.
2. Divide the total by the number of units.
3. Machines: start from the end and undo each step. Check by going forward.

## Opposites

- + and – undo each other.
- × and ÷ undo each other.
- "4 times as many" is 4 units.

## Your turn

Answer at the back of the book.

**Try** **A shop has 114 pens and rulers. The number of pens is 5 times the number of rulers. How many rulers are there?** ☐

## Start here

Do these first. Take your time.

**1** **Find the value of each.**

**(a)** 3706 × 4 = ☐

**(b)** 85 × 47 = ☐

**(c)** 7524 ÷ 6 = ☐

**(d)** 8150 ÷ 5 = ☐

Working: _____

_____

**2** **Round each number to the nearest 1000. Then estimate the answer.**

**(a)** 4986 × 4 is about ☐

**(b)** 9120 ÷ 3 is about ☐

**(c)** 7020 – 2980 is about ☐

**3** **A hall has 28 rows of 36 chairs. Then 45 chairs are broken and taken away. How many chairs are left in the hall?** ☐ **chairs**

Working: _____

_____

**4** **A baker packs 5468 buns into boxes of 8.**

**(a)** How many full boxes does she fill? ☐ boxes

**(b)** How many buns are left over? ☐ buns

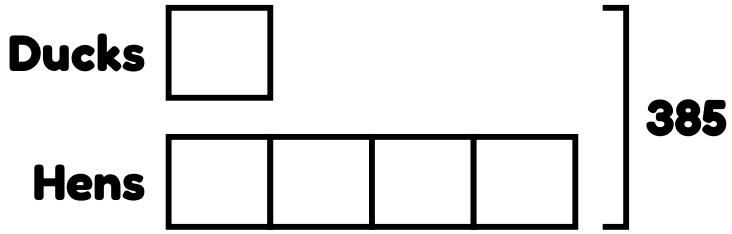
Working: _____

## Start here

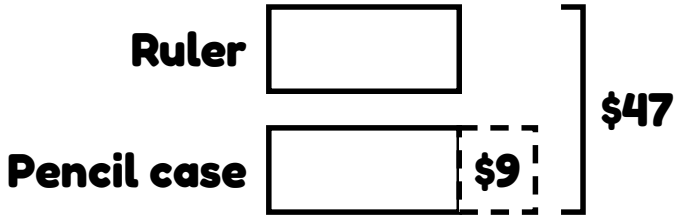
Keep going. Check your work.

**5** A farm has 4 times as many hens as ducks. There are 385 hens and ducks in all.

How many ducks are there? ☐ ducks

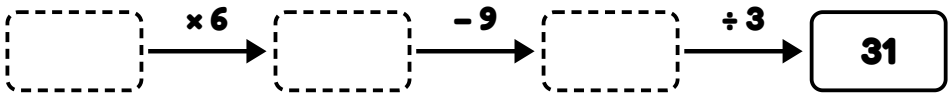



**6** A ruler and a pencil case cost $47 together. The pencil case costs $9 more than the ruler.




(a) The ruler costs $ ☐ (b) The pencil case costs $ ☐

**7** Mei picks a secret number. She multiplies it by 6, takes away 9, then divides by 3. She gets 31. What is her number? ☐




**8** A number trick. Start with a number. Multiply by 3, add 12, then divide by 3. Last, subtract the number you started with.

(a) Start with 6. You end with ☐

(b) Start with 15. You end with ☐

(c) Whatever you start with, you end with ☐

## Start here

Nearly there. You can do it.

**9** **Find the missing numbers.**

**(a)** 7 × ☐ = 4249

**(b)** ☐ ÷ 9 = 124

**(c)** 5000 − ☐ = 2738

**(d)** ☐ + 1905 = 6400

**10** **Mrs Lee buys 6 notebooks at $3 each and 4 pens at $2 each. She pays with $50. What is her change? $** ☐

Working: _____

_____

**11** **In a chain, add the last 2 numbers to make the next number. Fill in the missing numbers.**

**(a)** 4, 9, ☐, 22, ☐

**(b)** ☐, 6, 11, ☐, 28

Working: _____

**12** **Kumar picked 3 times as many mangoes as Aisha. They picked 480 mangoes in all. How many more mangoes did Kumar pick than Aisha?** ☐ **mangoes**

Draw a bar model here.

_____

# Think harder

Hints are upside down. Try first!

★ Beyond Level 4: Order of operations

**1** **Rule: do brackets first, then × and ÷, then + and –. Work from the left.**

**(a)** 60 – 4 × 9 + 12 ÷ 3 = ☐

**(b)** 5 × (13 + 7) – 6 = ☐

Working: _____

_____

**Hint.** In part (a), do 4 × 9 and 12 ÷ 3 before anything else.

**2** **In a chain, add the last 2 numbers to make the next number. The 5th number is 41 and the 6th is 66. What is the 1st number?** ☐

Working: _____

_____

_____

**Hint.** Work backwards. The 4th number is 66 – 41.

**3** **Mei has 5 times as many stickers as Lily. Mei gives 36 of hers to Lily. Then both girls have equal amounts.**

**(a)** Lily at first: ☐ stickers

**(b)** Mei at first: ☐ stickers

Draw a bar model here.

_____

_____

**Hint.** Before the gift, how many more does Mei have than Lily?

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 4: Signs and brackets

**1** **Rule: brackets first, then × and ÷, then + and –.**

**(a)** Write + – × or ÷ in each circle: 9 ○ 4 ○ 6 = 30

**(b)** Put ONE pair of brackets in 7 + 3 × 8 – 4 to make 76. Write the numbers inside the brackets: ☐ + ☐

Try: _____

**Hint.** For (a), estimate first. A big answer needs ×.

**2** **Ben and Mei both save for a bike. Plan A: start with $15 and save $8 each week. Plan B: start with $0 and save $13 each week.**

| Week | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Plan A | $15 | | | | |
| Plan B | $0 | | | | |

**(a)** In which week do both plans have the same amount? Week ☐

**(b)** How much is that? $ ☐

**Hint.** Fill in the table. Look at the gap between the plans each week.

**3** **Fill in the missing digits.**
**3 □ 4 × 7 = 2 □ 3 8**

**(a)** The missing digit in 3 □ 4 is ☐

**(b)** The missing digit in 2 □ 3 8 is ☐

Try: _____

_____

**Hint.** Start with the ones: what is 4 × 7?

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Find the value.** **4728 × 5 =** ☐ **9135 ÷ 7 =** ☐

Working: _____

**2** **Ali saves $15 a week for 12 weeks. Then he spends $64. How much money does he have left? $** ☐

Working: _____

**3** **There are 4 times as many red cars as blue cars. There are 275 cars in all. How many red cars are there?** ☐

Working: _____

**4** **I think of a number. I add 17, multiply by 3 and take away 20. The answer is 94. What is my number?** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

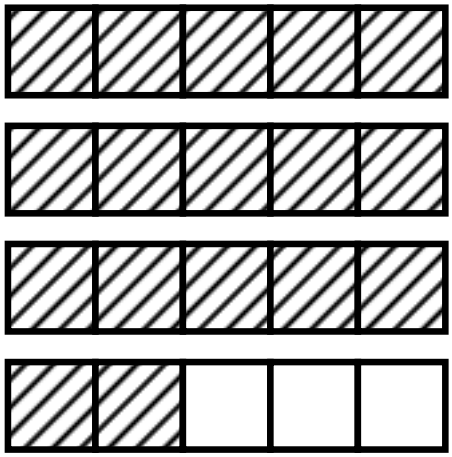
# Learn it: Fractions

**My goal.** I can change between improper fractions and mixed numbers. I can find a fraction of a set and compare fractions with different denominators.

A fraction bar shows equal parts of one whole. A fraction of a set uses a bar model: divide the total by the denominator to find 1 unit.

## Worked example 1

**Write 17/5 as a mixed number.**

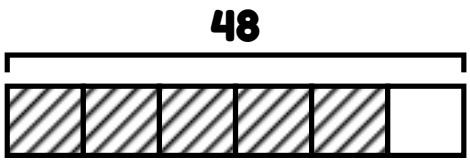


1. 5 fifths make 1 whole. Divide: 17 ÷ 5 = 3 with 2 left.
2. There are 3 wholes and 2 fifths.

**Answer: 17/5 = 3 2/5.**

## Worked example 2

**There are 48 stickers. 5/6 of them are gold. How many are gold?**




1. 6 units = 48, so 1 unit = 48 ÷ 6 = 8.
2. Gold is 5 units: 5 × 8 = 40.

**Answer: 40 stickers are gold.**

## How to do it

1. Improper to mixed: divide the top by the bottom.
2. Mixed to improper: whole × bottom + top, over the bottom.
3. Fraction of a set: 1 unit is the total ÷ the bottom number.

## Comparing

- Same bottom: the greater top is more.
- Same top: the smaller bottom is more.
- Or compare with 1/2.
- Equivalent: multiply or divide top and bottom by the same number.
- Whole from a part: 3 units = 36, so 1 unit = 12.

## Your turn

Answer at the back of the book.

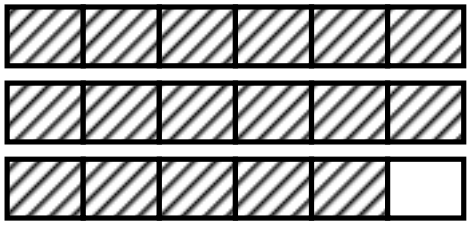
Try **Write 3 4/5 as an improper fraction.** ☐ / ☐

Name _____

## Start here

Do these first. Take your time.

**1** **Each bar is 1 whole cut into sixths. Look at the shaded parts.**



**(a)** How many sixths are shaded? ☐ sixths

**(b)** Write it as a mixed number. ☐ ☐ / ☐

**2** **Write each as a mixed number.**

**(a)** 19/4 = ☐ ☐ / ☐

**(b)** 23/6 = ☐ ☐ / ☐

**(c)** 31/8 = ☐ ☐ / ☐

**(d)** 40/5 = ☐

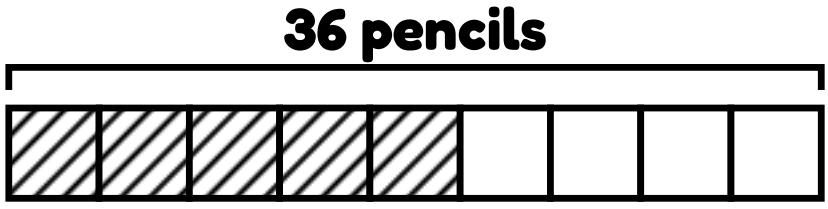
**3** **Write each as an improper fraction.**

**(a)** 3 2/5 = ☐ / ☐

**(b)** 5 1/3 = ☐ / ☐

**(c)** 2 7/8 = ☐ / ☐

**4** **A box has 36 pencils. 5/9 of them are red.**



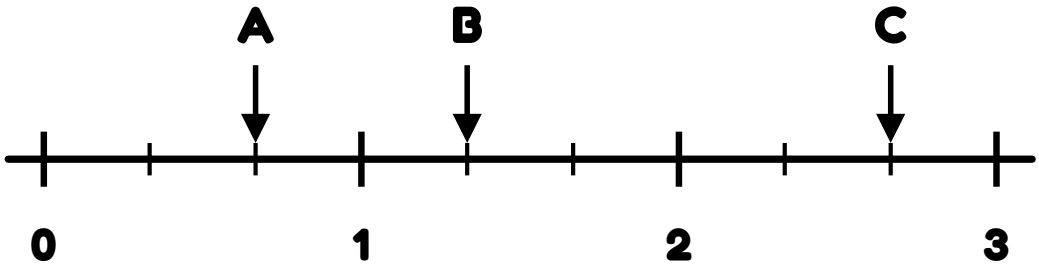

**(a)** How many pencils are red? ☐ pencils

**(b)** How many are not red? ☐ pencils

## Start here

Keep going. Check your work.

**5** **The number line is marked in thirds. Write the number each arrow shows.**




**(a)** A = ☐ / ☐ **(b)** B = ☐ ☐ / ☐

**(c)** C as an improper fraction = ☐ / ☐

**6** **Mei has read 3/7 of her book. She has read 36 pages.**

**(a)** Pages in the whole book: ☐

**(b)** How many pages are left to read? ☐ pages

Draw a bar model here.

**7** **Circle the correct sign.**

**(a)** 3/4 > < 5/8 **(b)** 4/6 > < 7/12 **(c)** 3/8 > < 5/9

**(d)** 5/6 > < 5/8

**8** **Write the fractions in order. Begin with the smallest.**

**3/4 2/3 5/12 1/2**

_____

## Start here

Nearly there. You can do it.

**9** **Fill in the missing numbers.**

**(a)** 2/3 = ☐/12 **(b)** 15/20 = 3/☐ **(c)** 9/12 = ☐/4

**(d)** 4/5 = 12/☐

**10** **Each pizza is cut into 8 slices. The children eat 27 slices.**

**(a)** How many pizzas is that? ☐ ☐/☐ pizzas

**(b)** How many slices are in 5 whole pizzas? ☐ slices

**(c)** How many more slices make 4 whole pizzas? ☐ slices

**11** **A rope is 2 2/3 m long. It is cut into pieces that are each 1/3 m long.**

**(a)** How many pieces are there? ☐ pieces

**(b)** Siti uses 5 pieces. How much rope is left? ☐ m

Working: _____

**12** **Aisha has 48 stickers. Mei gets 1/4 of the stickers. Ben gets 1/3 of the stickers that are left. How many stickers does Aisha keep?** ☐

Working: _____

_____

**13** **How many?**

**(a)** quarters in 5 wholes: ☐

**(b)** sixths in 3 wholes: ☐

**(c)** halves in 2 1/2: ☐

## Think harder

Hints are upside down. Try first!

**1** **Find the fraction. It has 7 on the bottom. It lies between 3 and 4, closer to 4 than to 3. Its top number is even.**

[ ] / [ ]

List: _____

_____

_____

**Hint.** List the fractions between 3 and 4. Cross out those that do not fit.

**2** **A book took 2/5 of Mei's money. A toy took 1/3 of her money. She has $24 left.**

**How much money did Mei start with? $** [ ]

Draw a bar model here.

_____

_____

_____

**Hint.** Which number do 5 and 3 both go into?

**3** **Siti reads 1/3 of a book on Monday, then 3/4 of the pages that remain on Tuesday. 20 pages are still unread. How long is the book?** [ ] **pages**

Draw a bar model here.

_____

_____

**Hint.** After Monday, 2/3 of the book is left. 1/4 of that is 20 pages.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Think about fractions with denominator 20 that are greater than 1/4 and less than 3/5.**

**(a)** How many are there? ☐

**(b)** Which one is equal to 1/2? ☐ /20

**Hint.** Write 1/4 and 3/5 as twentieths. List the ones in between.

★ Beyond Level 4: Unit fractions

**2** **Write each fraction as a sum of 2 different unit fractions. A unit fraction has 1 on top.**

**(a)** 5/6 = 1/☐ + 1/☐

**(b)** 3/4 = 1/☐ + 1/☐

**(c)** 7/10 = 1/☐ + 1/☐

**Hint.** Try 1/2 first. What is left?

**3** **In a class, 2/5 of the children are girls. 1/3 of the girls wear glasses. 6 girls wear glasses. How many children are in the class?** ☐

Draw a bar model here.

**Hint.** Find the number of girls first.

**4** **Compare each pair. Circle the correct sign.**

**(a)** 7/9 > < 8/10

**(b)** 11/12 > < 9/10

**(c)** 14/15 > < 19/20

**Hint.** How far is each fraction from 1?

Name _____

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Write 29/6 as a mixed number.** ☐ ☐ / ☐

Working: _____

**2** **Write 5 3/8 as an improper fraction.** ☐ / ☐

Working: _____

**3** **Which is greater, 5/8 or 7/12? Circle it.** **5/8** **7/12**

Working: _____

**4** **4/9 of the beads in a bag are blue. There are 28 blue beads. How many beads are in the bag?** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

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Something I want to ask or try: _____

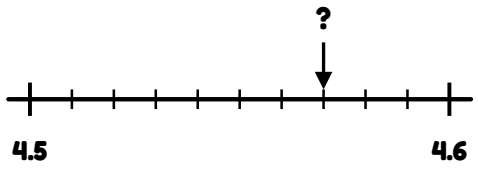
# Learn it: Decimals

**My goal.** I can read decimals on number lines, compare and order them and round them. I can solve money and mass puzzles with bar models.

On a number line, first find what one small step is worth. To compare decimals, line up the decimal points and compare place by place from the left.

## Worked example 1

**What number does the arrow show?**

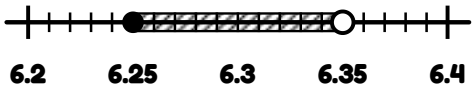



1. 4.5 to 4.6 has 10 steps. 1 step = 0.01.
2. The arrow is 7 steps after 4.5.

**Answer: The arrow shows 4.57.**

## Worked example 2

**A number has 2 decimal places. It rounds to 6.3 to 1 decimal place. What are the least and greatest it can be?**




1. 6.25 is halfway, so it rounds up to 6.3.
2. 6.35 rounds up to 6.4, so the last number is 6.34.

**Answer: Least 6.25 and greatest 6.34.**

## How to do it

1. Number line: find what 1 small step is worth.
2. Compare: line up the points. Add zeros to fill the gaps.
3. Round: find the 2 neighbours. Halfway rounds up.

## Place value

| ones | . | tenths | hundredths | thousandths |
|---|---|---|---|---|
| 3 | . | 4 | 0 | 5 |

- 3.405 has 4 tenths, 0 hundredths and 5 thousandths.

## Your turn

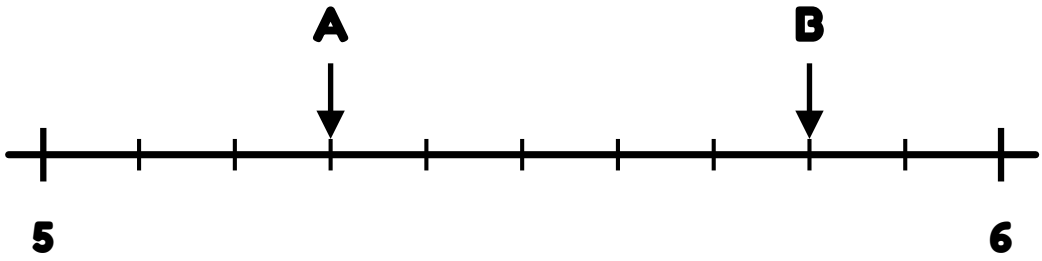
Answer at the back of the book.

Try **Round 8.46 to 1 decimal place.** ☐

## Start here

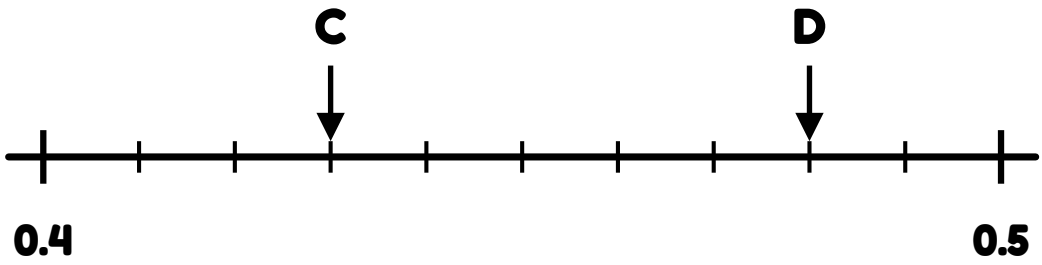
Do these first. Take your time.

**1** **Each small step on the number line is the same size. Write the number each arrow shows.**




(a) A = ☐ (b) B = ☐

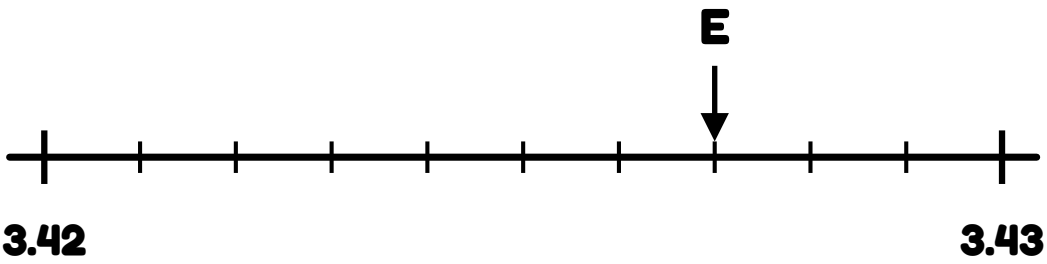
**2** **Now each small step is 0.01. Write the number each arrow shows.**




(a) C = ☐ (b) D = ☐

(c) Which arrow shows a number that rounds to 0.5? C D

**3** **Each small step is 0.001. Write the number the arrow shows.**




(a) E = ☐

## Start here

Keep going. Check your work.

**4** **Circle the correct sign.**

**(a)** 4.06 > < = 4.6

**(b)** 0.38 > < = 0.308

**(c)** 12.5 > < = 12.50

**(d)** 7.09 > < = 7.1

**5** **Write the numbers in order. Begin with the smallest.**

**5.205 5.25 5.052 5.5 5.025**

_____

**6** **Write the value of the digit 6 in each number.**

**(a)** 4.562: ☐ **(b)** 6.043: ☐ **(c)** 0.936: ☐

**7** **Round 6.847.**

**(a)** Nearest 1: ☐

**(b)** Nearest 0.1: ☐

**(c)** Nearest 0.01: ☐

Working: _____

**8** **Ali buys a ruler for $1.85 and a notebook for $2.40. He pays with a $5 note. How much change does he get? $** ☐

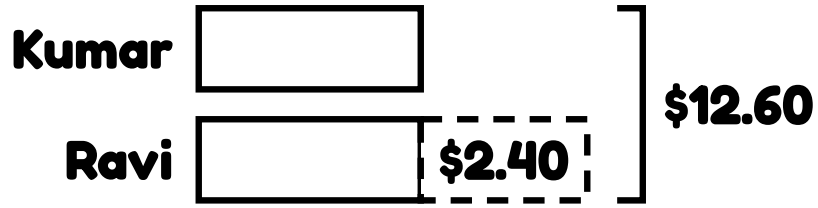
Working: _____

_____

## Start here

Nearly there. You can do it.

**9** **Ravi and Kumar have $12.60 altogether. Ravi has $2.40 more than Kumar.**




**(a)** Kumar has $ ☐ **(b)** Ravi has $ ☐

**10** **A box holds 4 packs of rice. All the packs have the same mass. The box and packs together have a mass of 4.1 kg. The empty box has a mass of 0.5 kg.**

**(a)** The 4 packs together: ☐ kg

**(b)** One pack: ☐ kg

**(c)** Kumar takes out 1 pack. The box and packs now have a mass of ☐ kg

Working: _____

**11** **Find the value of each.**

**(a)** 4.7 + 2.35 = ☐

**(b)** 9.2 – 3.64 = ☐

**(c)** 2.5 × 4 = ☐

**(d)** 7.2 ÷ 6 = ☐

Working: _____

**12** **Write each as a decimal.**

**(a)** 7/10 = ☐ **(b)** 19/100 = ☐ **(c)** 3 1/2 = ☐

**(d)** 2 3/4 = ☐

## Think harder

Hints are upside down. Try first!

**1** **A number has 3 decimal places. Rounded to 2 decimal places it is 4.27.**

**(a)** The least it can be: ☐

**(b)** The greatest it can be: ☐

Working: _____

_____

**Hint.** Find the halfway marks next to 4.27. Which ones round to 4.27?

**2** **Use the digit cards 1, 4, 6 and 9 and a decimal point. Use every card once. The point goes between 2 digits. Make the number that is closest to 20.**

☐

List: _____

_____

_____

**Hint.** List the numbers that start with 1 and with 9. Which are near 20?

**3** **A mug costs $2.40 more than a cup. 3 mugs and 2 cups cost $17.20 altogether.**

**(a)** A cup costs $ ☐

**(b)** A mug costs $ ☐

Draw a bar model here.

_____

_____

**Hint.** Change the 3 mugs into cups. How much extra is that?

## ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **A full bottle of juice has a mass of 1.26 kg. When one third of the juice is left, the bottle has a mass of 0.58 kg. Find the mass of the bottle when it is empty.**

☐ kg

Working: _____

_____

**Hint.** The juice drunk is 2/3 of the juice. Its mass is 1.26 − 0.58.

**2** **Mei has 12 coins. They are all 10¢ coins and 50¢ coins. Altogether they are worth $3.60. How many 50¢ coins does she have?** ☐

Working: _____

_____

**Hint.** Suppose all 12 coins are 10¢. How much money is that?

**3** **Four children share $20.40. Each child gets $0.60 more than the one before. The youngest gets the least.**

**(a)** The youngest gets $ ☐

**(b)** The oldest gets $ ☐

Working: _____

**Hint.** Take away the extras first: 0.60, 1.20 and 1.80.

**4** **A pattern starts 0.35, 0.40, 0.45, 0.50 and goes on in the same way.**

**(a)** The 20th number is ☐

**(b)** Which place in the pattern is 2.05? Place ☐

Working: _____

**Hint.** Each place adds 0.05. How many steps after the 1st?

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Write the numbers in order. Begin with the smallest.**
**6.07 6.7 6.007 6.17**

_____

Working: _____

**2** **Round 7.846 to the nearest whole number ☐, 1 decimal place ☐, 2 decimal places ☐**

Working: _____

**3** **Siti buys 4 notebooks at $1.35 each and pays $10. Her change is $ ☐**

Working: _____

**4** **Rounding to 1 decimal place gives 3.6. The number has 2 decimal places. The least is ☐ and the greatest is ☐.**

Working: _____

**My score: ☐ out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

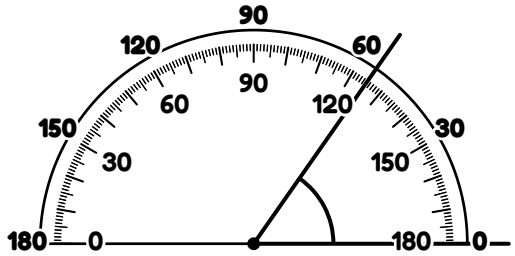
# Learn it: Angles

**My goal.** I can measure angles and find missing angles. I can find the angle between clock hands and turn on the 8-point compass.

You need a 180° protractor and a ruler for this unit. A full turn is 360°, a half turn is 180° and a quarter turn is 90°. To measure, count up from the 0 on one arm.

## Worked example 1

**How big is the angle? Use the scale that starts at 0 on the arm.**

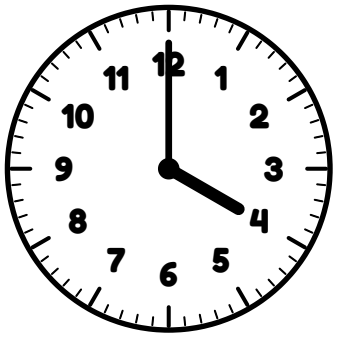



1. **Put the centre of the protractor on the corner. Put the 0 line on one arm.**
2. **Read the scale that starts at 0 on that arm.**

**Answer: The angle is 55°.**

## Worked example 2

**What is the angle between the clock hands at 4 o'clock?**




1. **A clock has 12 equal spaces in 360°. 360 ÷ 12 = 30, so 1 space is 30°.**
2. **At 4 o'clock the hands are 4 spaces apart: 4 × 30°.**

**Answer: The angle is 120°.**

## How to measure

1. Estimate first. Is it less than 90° or more?
2. Centre on the corner. 0 line on one arm.
3. Read from the 0 on that arm. Check with your estimate.

## Good to know

- Compass: 1 step = 45°.
- Clock: 1 space = 30°.
- Minute hand: 1 minute = 6°.
- ∠AOC is the angle at O between OA and OC.

## Your turn

Answer at the back of the book.

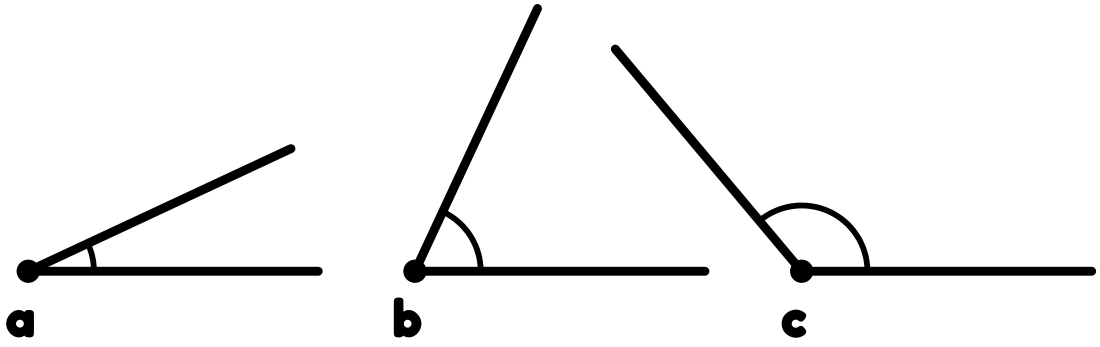
**Try** **Ali faces NE. He turns 135° clockwise. Which way does he face now?**

E SE S SW

## Start here

Do these first. Take your time.

**1** **Estimate each angle. Then measure it with your protractor. Use a ruler to make the arms longer first.**



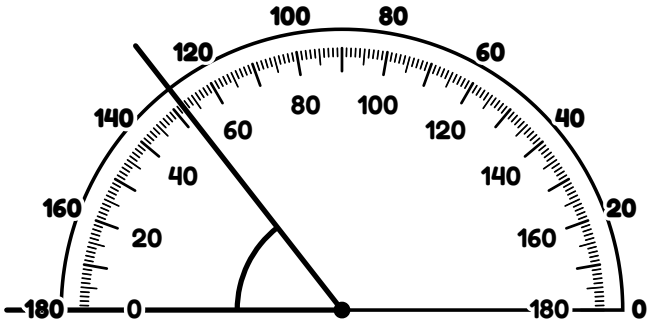

(a) a = ☐ ° (b) b = ☐ ° (c) c = ☐ °

**2** **Read the angle from the protractor.**

(a) The angle is ☐ °

(b) Is it smaller or greater than a right angle?

smaller greater



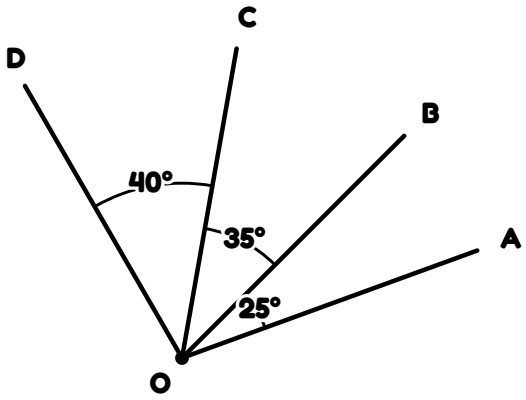

**3** **Rays OA, OB, OC and OD leave O. The 3 small angles lie side by side. Find the angles.**

(a) ∠AOC = ☐ °

(b) ∠BOD = ☐ °

(c) ∠AOD = ☐ °

(d) How many angles are there in all? ☐



## Start here

Keep going. Check your work.

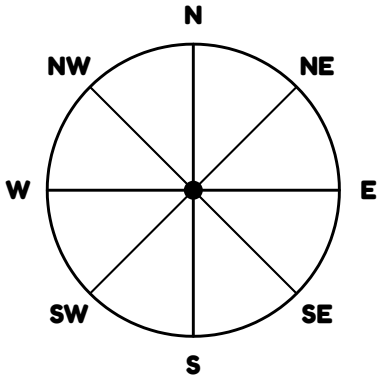
**4** **Each step round the compass is 45°. Find the direction.**

**(a)** Ali faces W. He turns 90° clockwise. He now faces ☐

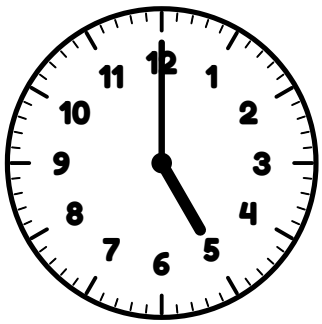
**(b)** Mei faces SE. She turns 180°. She now faces ☐

**(c)** Siti faces NE. She turns 270° clockwise. She now faces ☐

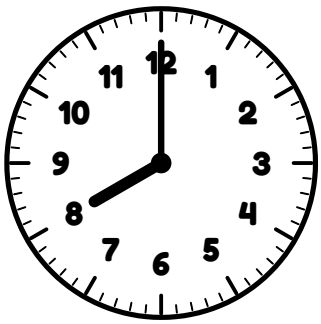
**(d)** Ravi turns 135° anticlockwise and now faces S. At first he faced ☐



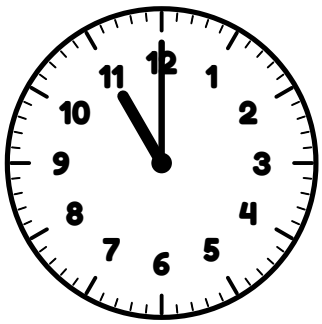

**5** **Find the smaller angle between the clock hands.**



**5 o'clock**



**8 o'clock**



**11 o'clock**

**(a)** 5 o'clock: ☐ °

**(b)** 8 o'clock: ☐ °

**(c)** 11 o'clock: ☐ °

**6** **Ben faces W. He wants to face NE. What is his smallest turn?**

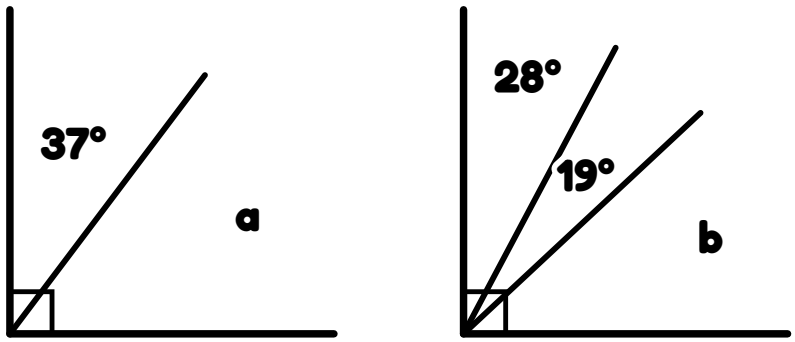
**(a)** The turn is ☐ °

**(b)** The direction is clockwise anticlockwise

## Start here

Nearly there. You can do it.

**7** **Each rectangle corner is 90°. Lines split the corners. Find the angles a and b. The drawings are not to scale.**




**(a)** a = ☐ ° **(b)** b = ☐ °

**8** **A robot faces E. It turns 135° anticlockwise. Then it turns 90° clockwise. Then it turns 45° clockwise.**

**(a)** It now faces ☐

**(b)** The turns add up to ☐ °

**9** **A corner of a rectangle is split into 2 angles. One angle is 5 times the other.**

**(a)** The smaller angle is ☐ °

**(b)** The greater angle is ☐ °

Draw a bar model here.

**10** **How long does the minute hand take to turn through each angle?**

**(a)** 90°: ☐ minutes **(b)** 180°: ☐ minutes

**(c)** 270°: ☐ minutes **(d)** 60°: ☐ minutes

## Think harder

Hints are upside down. Try first!

★ Beyond Level 4: Angles on a line and at a point

**1** **Angles on a straight line add up to 180°. Angles around a point add up to 360°.**

**(a)** On a straight line the angles are 47°, 85° and a. a = ☐ °

**(b)** Around a point the angles are 95°, 120°, 78° and b. b = ☐ °

Working: _____

_____

**Hint.** Add the angles you know. Then take the sum away from the total.

**2** **Find the smaller angle between the clock hands.**

**(a)** At 7:30 it is ☐ °

**(b)** At 1:30 it is ☐ °

Working: _____

_____

_____

**Hint.** At half past, the hour hand is halfway between 2 numbers.

**3** **A robot faces N. It turns 90° clockwise, then 135° anticlockwise, then 225° clockwise, then 45° anticlockwise.**

**(a)** It now faces ☐

**(b)** The turns add up to ☐ °

Working: _____

_____

**Hint.** Count the steps of 45°. Clockwise is plus, anticlockwise is minus.

## ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Six rays leave one point. They make 5 equal angles in a row. The 2 outside rays are 150° apart.**

**(a)** How many angles can you find in all? ☐

**(b)** Count the right angles among them. ☐

Working: _____

**Hint.** One small angle is 30°. Count 1-part, 2-part, 3-part angles.

**2** **A robot faces N. It turns 135° clockwise again and again.**

**(a)** How many turns until it faces N again? ☐

**(b)** After 50 turns it faces ☐

List: _____

**Hint.** 135° is 3 steps of 45°. Write the directions after each turn.

**3** **A rectangle corner of 90° is cut by 2 lines. The middle angle is 2 times the smallest. The greatest is 6 times the smallest.**

**(a)** The smallest angle is ☐ °

**(b)** The greatest angle is ☐ °

Working: _____

**Hint.** Count the equal parts. 1 + 2 + 6 = 9.

**4** **Think about the smaller angle between the clock hands at each o'clock time. Which times give an angle between 100° and 160°?**

**(a)** How many times are there? ☐

**(b)** Write the times: ☐ ☐ ☐ ☐

**Hint.** Write the angle for every o'clock time first.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Find the smaller angle between the clock hands at 7 o'clock.** ☐°

Working: _____

**2** **Aisha faces S and turns 135° clockwise. Which way does she face now?** ☐

Working: _____

**3** **A corner of a square is split into 2 angles. One angle is 41°. How big is the other?** ☐°

Working: _____

**4** **The minute hand moves from 5:10 to 5:35. How many degrees does it turn?** ☐°

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

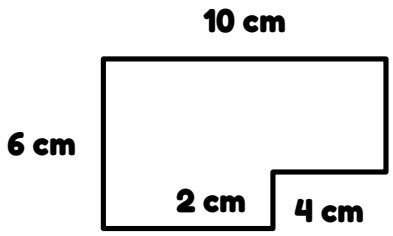
# Learn it: Area and perimeter

**My goal. I can find the area and the perimeter of shapes with square corners. I can compare rectangles with the same area.**

Area is the space inside a shape, in cm². Perimeter is the distance round the edge, in cm. For a shape with square corners, take a piece away from a big rectangle, or split it into rectangles.

## Worked example 1

**The shape has square corners. Find its area and its perimeter.**




1. Big rectangle: 10 × 6 = 60. The missing corner is 4 × 2 = 8. Area = 60 − 8 = 52.
2. Push the steps out to make a rectangle. The perimeter is 10 + 6 + 10 + 6 = 32.

**Answer: Area 52 cm². Perimeter 32 cm.**

## Worked example 2

**Rectangles with whole-cm sides have an area of 18 cm². Find the perimeter of each one.**

| Length | Width | Perimeter |
|---|---|---|
| 18 | 1 | 38 |
| 9 | 2 | 22 |
| 6 | 3 | 18 |

1. List the pairs that multiply to 18 in order: 1 × 18, 2 × 9, 3 × 6.
2. Perimeter = 2 × (length + width).

**Answer: The smallest perimeter is 18 cm, from 6 cm by 3 cm.**

## How to do it

1. Missing sides: use the opposite sides.
2. Area: split into rectangles, or take a piece away.
3. Perimeter: push the steps out to a rectangle.

## Good to know

- Rectangle: area = length × width.
- Perimeter = 2 × (length + width).
- A square has 4 equal sides.

## Your turn

Answer at the back of the book.

**Try** A rectangle is 12 cm long and 5 cm wide. A 3 cm by 2 cm corner is cut off. What is the area of the shape that is left? ☐ cm²

## Start here

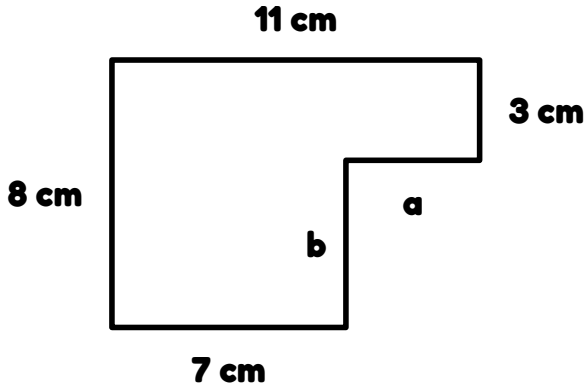
Do these first. Take your time.

**1** **The shape has square corners. Find the missing sides a and b. Then find the area and the perimeter.**

(a) a = ☐ cm b = ☐ cm

(b) Area: ☐ cm²

(c) Perimeter: ☐ cm

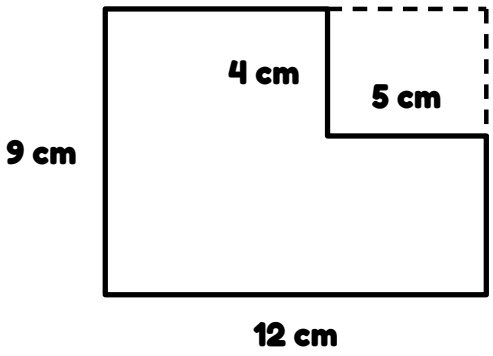



**2** **A 12 cm by 9 cm rectangle has a corner cut off. The corner is shown with a dashed line.**

(a) Area of the whole rectangle: ☐ cm²

(b) Area of the corner cut off: ☐ cm²

(c) Area of the shape left: ☐ cm²




**3** **Every rectangle here has whole-cm sides and covers 28 cm². Fill in the table. The length is the greater side.**

(a) Fill in the table.

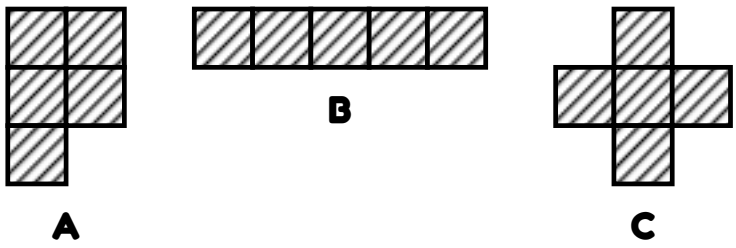
(b) The least perimeter is ☐ cm

| Length | Width | Perimeter |
|---|---|---|
| 28 | | |
| 14 | | |
| 7 | | |

## Start here

Keep going. Check your work.

**4** **Each shape is made of 5 square tiles that are each 1 cm by 1 cm. The tiles touch along whole sides.**




**(a)** The area of each shape: ☐ $cm^2$

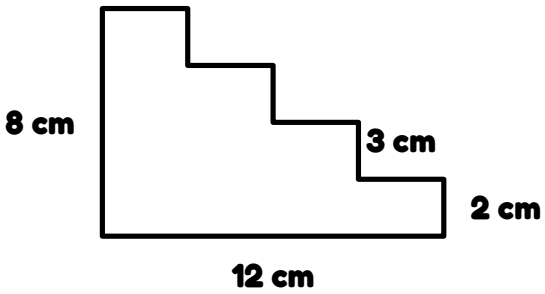
**(b)** The perimeters of A, B and C: ☐ cm, ☐ cm, ☐ cm

**(c)** The shape with the least perimeter is A B C

**5** **The staircase has 4 equal steps. Each step is 3 cm across and 2 cm high.**

**(a)** Perimeter: ☐ cm

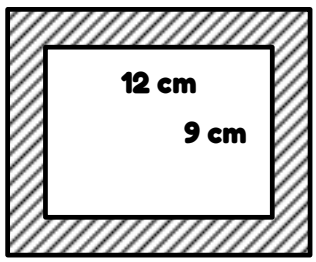
**(b)** Area: ☐ $cm^2$




**6** **A photo is 12 cm long and 9 cm wide. It sits in a frame that is 2 cm wide all round.**

**(a)** Photo and frame: ☐ cm by ☐ cm

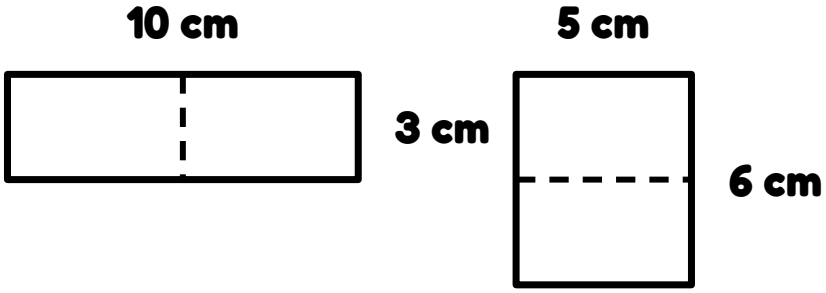
**(b)** Area of the frame only: ☐ $cm^2$



## Start here

Nearly there. You can do it.

**7** **Cut a 10 cm by 3 cm strip on the dashed line. Join the 2 pieces to make a 5 cm by 6 cm rectangle.**




**(a)** The area of the strip is ☐ $cm^2$ and the area of the rectangle is ☐ $cm^2$.

**(b)** The perimeter of the strip is ☐ cm and of the rectangle is ☐ cm.

**8** **A rectangle has an area of 72 $cm^2$. Its length is 9 cm.**

**(a)** Its width is ☐ cm

**(b)** Its perimeter is ☐ cm

Working: _____

**9** **A hall floor is 12 m long and 8 m wide. It is covered with square tiles that are each 2 m by 2 m.**

**(a)** How many tiles cover the floor? ☐ tiles

**(b)** The tiles along the edge are dark. How many tiles are dark? ☐ tiles

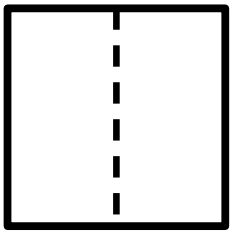
Working: _____

_____

## Think harder

Hints are upside down. Try first!

**1** **A square is cut into 2 identical rectangles, as shown. Each rectangle has a perimeter of 30 cm. The area of the square is ☐ cm².**



Working: _____

_____

**Hint.** Each rectangle has 3 sides that equal the side of the square.

**2** **Mei has 28 m of fence for a rectangular pen. The sides are whole numbers of metres.**

**(a)** The greatest area when the pen needs 4 fenced sides: ☐ m²

**(b)** The greatest area when one side is a wall that needs no fence: ☐ m²

List: _____

_____

_____

**Hint.** For (b), the fence makes 2 short sides and 1 long side.

**3** **A garden is 12 m long and 9 m wide. Two paths, each 2 m wide, cross it. One path runs along the length and one runs across the width. What is the area of the garden that is NOT path? ☐ m²**

Working: _____

_____

_____

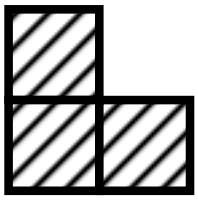
**Hint.** Slide the paths to the edge. The grass makes 1 rectangle.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1**

**Three identical squares are joined in an L shape. The perimeter of the L shape is 40 cm. Find its area.**



☐ **cm²**

**Hint.** Count the sides of squares round the edge of the L shape.

**2**

**Take a square with sides of 4 cm. Its area is 16 cm² and its perimeter is 16 cm. The numbers match! Find 1 other rectangle with whole-cm sides where they match.**

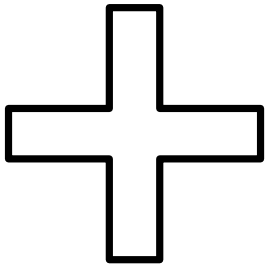
**Length** ☐ **cm** **Width** ☐ **cm**

Try: _____

**Hint.** Try a width of 1, then 2, then 3. Does the area equal the perimeter?

**3**

**Two strips, each 10 cm by 2 cm, cross in the middle. Together they make a plus shape.**



**(a)** Area of the plus shape: ☐ **cm²**

**(b)** Perimeter of the plus shape: ☐ **cm**

**Hint.** The middle square is part of both rectangles.

**4**

**A square floor is covered with square tiles. The tiles along the edge are dark and all the others are light.**

**(a)** The floor is 8 tiles by 8 tiles. How many dark tiles? ☐

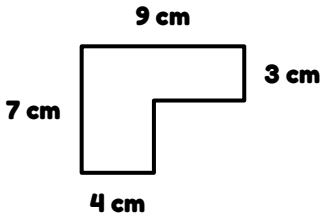
**(b)** A floor has 60 dark tiles. How many tiles long is the floor? ☐

**Hint.** Take the light block away from the whole floor.

## ✔ Check yourself

No hints. Try each one on your own.

**1** Find the area of the shape. ☐ cm²




**2** A rectangle has an area of 60 cm² and a length of 12 cm. Find its perimeter.

☐ cm

Working: _____

**3** A 10 cm by 7 cm photo has a frame 1 cm wide on every side. Find the area of the frame alone. ☐ cm²

Working: _____

**4** Every rectangle has whole-cm sides and covers 24 cm². Find the least perimeter any of them can have. ☐ cm

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

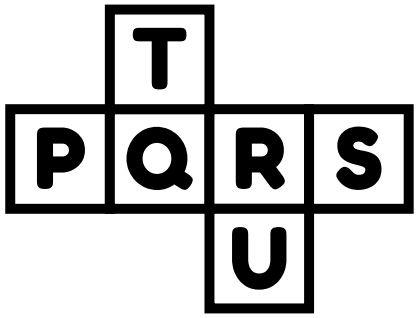
# Learn it: Symmetry and nets

**My goal.** I can tell which nets fold into a cube and which faces are opposite. I can complete a figure so that a line is a line of symmetry.

A net is a flat shape that folds into a solid. A cube net has 6 squares. In a line of 3 squares, the 2 end squares are opposite faces. On a dice, opposite faces add up to 7.

## Worked example 1

**Does this net fold into a cube? Which face is opposite P?**

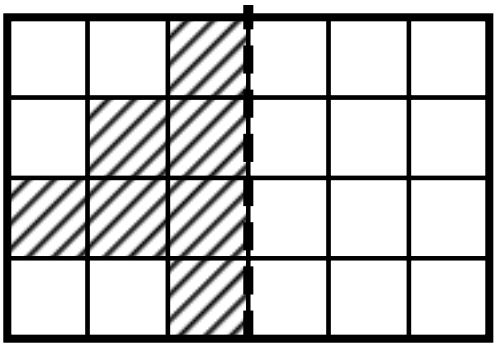



1. Choose a base. The row of 4 squares wraps round as walls.
2. One square is above the row and one is below. They make the lid and the base.
3. P and R have 1 square between them. They are opposite.

**Answer: It folds into a cube. R is opposite P.**

## Worked example 2

**Complete the figure. The dashed line must be a line of symmetry.**



1. Flip each shaded square across the line.
2. The partner is the same distance from the line.

**Answer: Each shaded square gets a partner the same distance from the line.**

## How to do it

1. Net: choose a base. Fold the squares round it in your head.
2. A net needs 1 base, 1 lid and 4 walls.
3. Symmetry: count squares from the line to the partner.

## Good to know

- Dice: opposite faces add up to 7.
- A rectangle has 2 lines of symmetry. A square has 4.
- No fold: 4 squares at one corner, or 5 in a row.

## Your turn

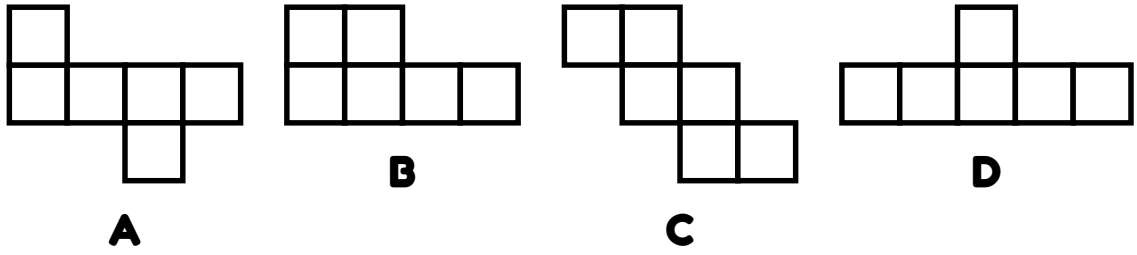
Answer at the back of the book.

Try **Opposite faces of a dice add up to 7. What number is opposite 4?** ☐

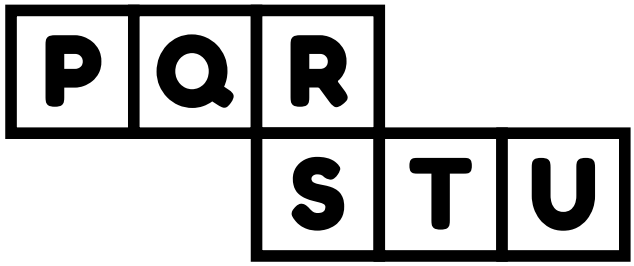
## Start here

Do these first. Take your time.

**1** **Which of these nets fold into a cube? Circle the letters.**



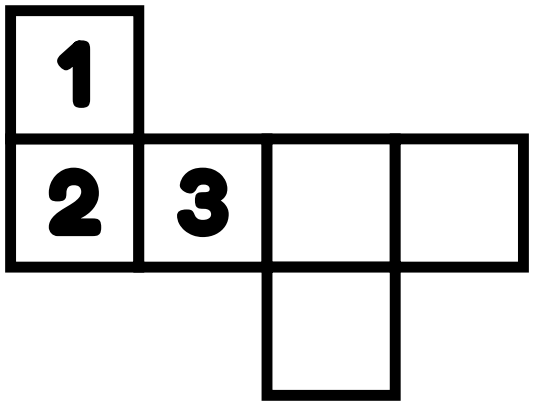

**2** **The net folds into a cube. Name the letter opposite each one.**




**(a)** Opposite P: ☐ **(b)** Opposite Q: ☐

**(c)** Opposite S: ☐

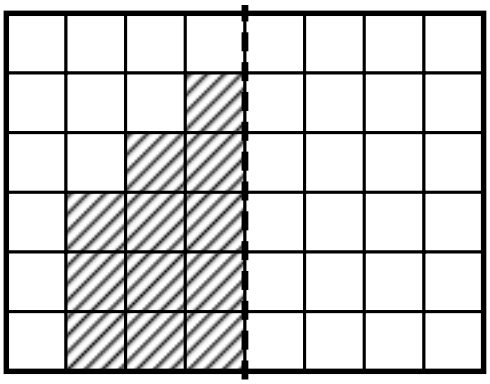
**3** **Each pair of opposite faces on a dice has a total of 7. Fill in the blank squares. Go row by row.**



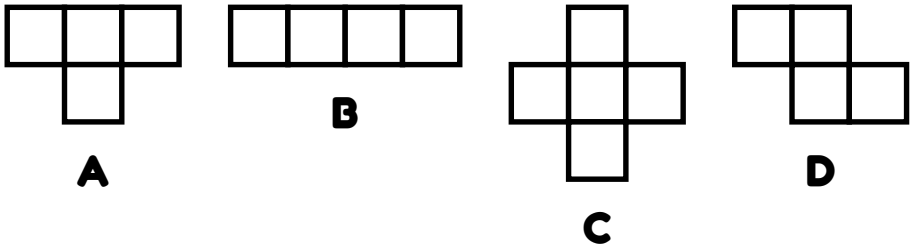
## Start here

Keep going. Check your work.

**4** **Fold on the dashed line. Shade squares so the 2 halves match. How many squares are shaded in all?** ☐



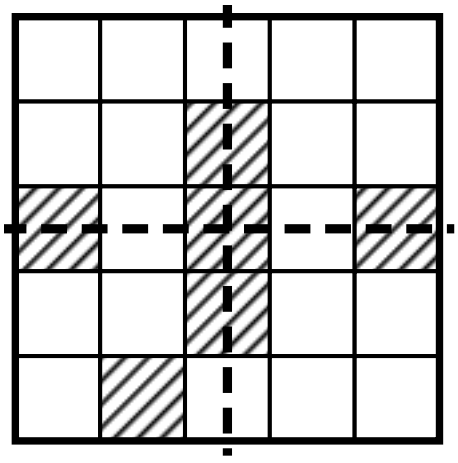
**5** **How many lines of symmetry does each shape have?**




**(a)** A: ☐ **(b)** B: ☐ **(c)** C: ☐ **(d)** D: ☐

**6** **There are 2 dashed lines. Shade 1 more square. The halves must then match on one of the lines.**

**(a)** How many different squares can you shade? ☐



## Start here

Nearly there. You can do it.

**7** **A dice sits on a table. The top face shows 6 and the face in front shows 4.**

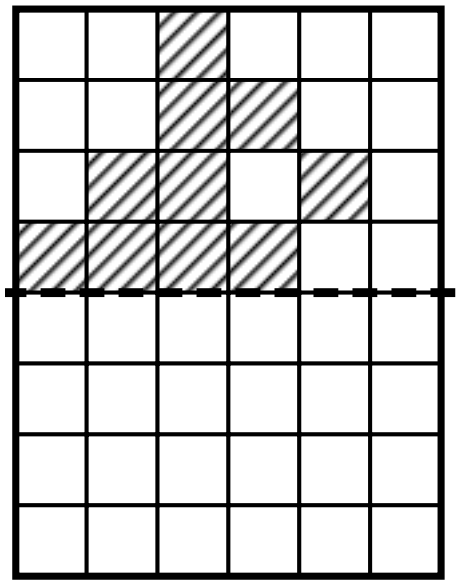
**(a)** The bottom face shows ☐

**(b)** The face at the back shows ☐

**(c)** The 4 side faces add up to ☐

**8** **A dice net is made of 6 squares. Each square is 3 cm by 3 cm. Find the area of the card.** ☐ **cm²**

**9** **Shade squares so the top and bottom halves match on the dashed line. How many are shaded in all?** ☐



**10** **A cube has edges of 5 cm.**

**(a)** All 12 edges are 5 cm long. The edges add up to ☐ cm

**(b)** The area of its net is ☐ cm²

## Think harder

Hints are upside down. Try first!

**1** **Mei paints each face of a cube red or blue.**

**(a)** **No 2 red faces may share an edge. The most red faces is** ☐

**(b)** **Now red faces may share an edge, but no 2 red faces may be opposite. The most red faces is** ☐

Working: _____

_____

**Hint.** Which faces share an edge with a given face?

**2** **Mei sees 3 faces of a dice at the same corner. Their numbers add up to 10.**

**(a)** **The numbers she can see, smallest first:** ☐ ☐ ☐

**(b)** **The numbers she cannot see, smallest first:** ☐ ☐ ☐

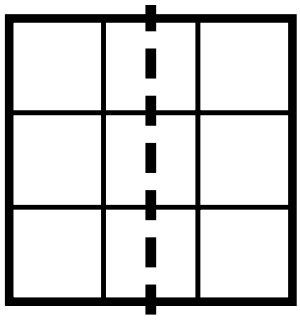
Working: _____

_____

**Hint.** At a corner, the 3 faces are from 3 different opposite pairs.

**3**

**On a 3 by 3 grid, shade exactly 3 squares. A fold down the middle must make the halves match. How many patterns can you make?** ☐



List: _____

_____

_____

**Hint.** Middle-column squares match themselves. The others come in pairs.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 4: Painted cubes

**1** **Paint all the outside faces of a big cube that is 3 by 3 by 3. Then cut it into 27 small cubes.**

**(a)** Small cubes with 3 painted faces: ☐

**(b)** Small cubes with 2 painted faces: ☐

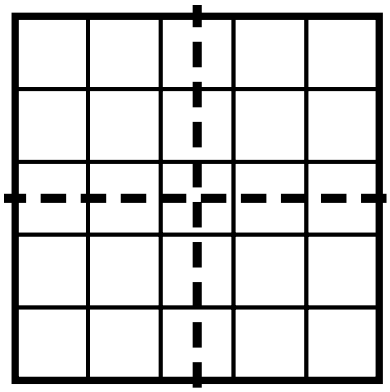
**(c)** Small cubes with 1 painted face: ☐

**(d)** Small cubes with no paint: ☐

**Hint.** Corners, edges and faces of the big cube. Count each kind.

**2**

**Shade just 1 square of a 5 by 5 grid. Some squares give a picture with a line of symmetry. The line can go up-down, across or along a diagonal.**



**(a)** The number of squares that give at least 1 line of symmetry: ☐

**(b)** The number that give exactly 1 line: ☐

**Hint.** Try a corner square, then an edge square, then the middle.

**3** **A dice has 1 on top, 2 facing north and 3 facing east. It rolls north, then east, then north again. A roll north puts the north face on the table. A roll east puts the east face on the table.**

**(a)** The top face is now ☐

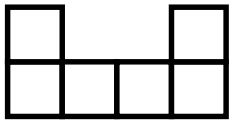
**(b)** The bottom face is now ☐

**Hint.** Write the top, north and east faces after each roll.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **Does this net fold into a cube?** **Yes** **No**



**2** **A rectangle has a different length and width. How many lines of symmetry does it have?** ☐

Working: _____

**3** **A dice has 2 on top and 3 in front. The bottom shows** ☐ **and the back shows** ☐**.**

Working: _____

**4** **A cube has edges of 4 cm. Its net has an area of** ☐ **cm².**

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

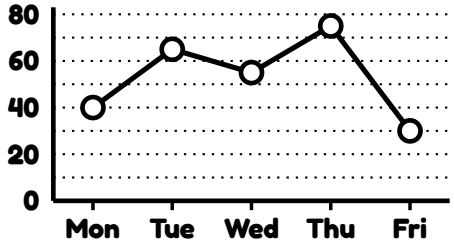
# Learn it: Tables and line graphs

**My goal. I can read values and changes on line graphs. I can find missing numbers in tables and spot graphs with scales that mislead.**

On a line graph, read the title and the scale first. The change between 2 neighbouring points is the difference of their values. A table can have a missing number: use the totals to find it.

## Worked example 1

**The graph shows the cups of barley water a stall sold. Each line is worth 10. How many cups were sold on Wednesday? When was the greatest drop?**




1. **The Wednesday point is halfway between 50 and 60: 55 cups.**
2. **Find each change. Thu to Fri is 75 – 30 = 45 down. That is the greatest drop.**

**Answer: 55 cups. The greatest drop was Thursday to Friday.**

## Worked example 2

**Find the missing number. Each row and each column adds to its total.**

| | Week 1 | Week 2 | Total |
|---|---|---|---|
| Ali | 14 | ? | 31 |
| Mei | 20 | 18 | 38 |
| Total | 34 | 35 | 69 |

1. **Use Ali's row: 14 + ? = 31, so ? = 31 – 14 = 17.**
2. **Check the Week 2 column: 17 + 18 = 35.**

**Answer: The missing number is 17.**

## How to read data

1. Read the title and the scale. What is one line worth?
2. Change = the difference between 2 neighbours.
3. Missing numbers: use the row or column total.

## Good to know

- The steepest line is the greatest change.
- Check where each scale starts.

## Your turn

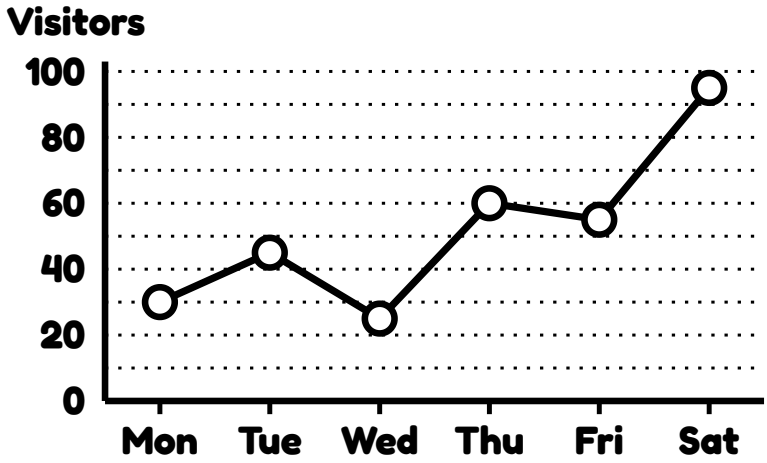
Answer at the back of the book.

**Try** **Each line on a graph is worth 5. Monday's point is at 70 books. Tuesday's point is 2 lines higher. How many books does Tuesday show?** ☐

## Start here

Do these first. Take your time.

**1** **The graph shows how many people visited a library each day. Each line is worth 10.**




**(a)** Visitors on Thursday: ☐ **(b)** Visitors on Tuesday: ☐

**(c)** Visitors on Monday and Tuesday together: ☐

**2** **Use the graph in Question 1. Find the greatest increase and the greatest decrease between 2 days.**

**(a)** Greatest increase: from ☐ to ☐, up by ☐

**(b)** Greatest decrease: from ☐ to ☐, down by ☐

Changes: _____

**3** **The table shows the library books 3 classes borrowed. Each row and each column adds up to its total. Write the missing numbers.**

**(a)** Fill in the 6 empty boxes. Go row by row.

**(b)** Which class borrowed fewer books in Week 2 than in Week 1?

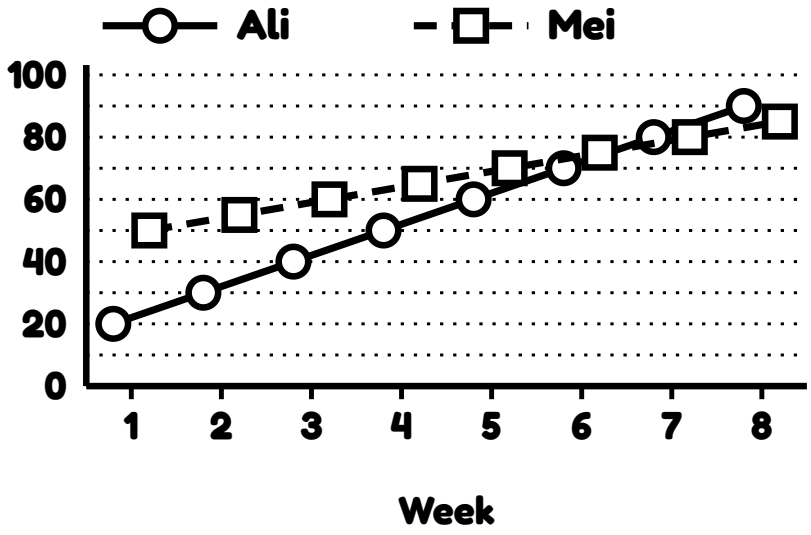
4A 4B 4C

| | Week 1 | Week 2 | Total |
|---|---|---|---|
| 4A | 38 | | 85 |
| 4B | | 29 | 70 |
| 4C | 33 | 41 | |
| Total | | | |

## Start here

Keep going. Check your work.

**4** **The graph shows money in Ali's box and in Mei's box. The amounts are at the end of each week.**




**(a)** Mei's money in Week 3: $ ☐ **(b)** Each week Ali adds $ ☐.

**(c)** Both have equal money in Week ☐.

**(d)** Ali first has more than Mei in Week ☐.

**5** **Siti reads more pages each day. She reads the same number more each day. Fill in the table.**

| Day | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Pages | 12 | | | 27 | |

**(a)** She reads ☐ more pages each day.

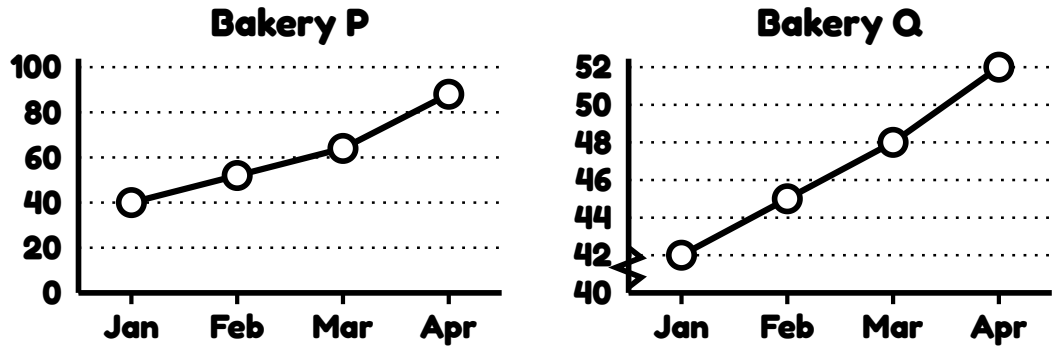
**(b)** In the 5 days she reads ☐ pages.

Working: _____

## Start here

Nearly there. You can do it.

**6** **Two bakeries show the cakes they sold. Look at each scale. Bakery Q's scale starts at 40.**




**(a)** From January to April, Bakery P went up by ☐ cakes.

**(b)** Bakery Q went up by ☐ cakes.

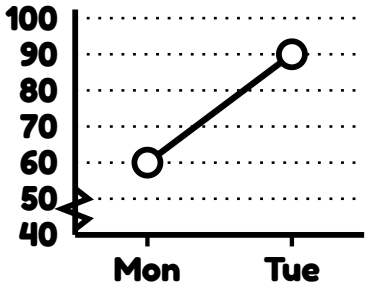
**(c)** Which bakery sold more extra cakes? P Q

**7** **The graph shows the pencils a shop sold. Look at the scale.**

**(a)** Monday: ☐ pencils

**(b)** Tuesday: ☐ pencils

**(c)** Tuesday's point is more than twice as high as Monday's. Is Tuesday's number more than double Monday's? Yes No




**8** **At 7 am the temperature was 25 °C. It rose by 2 °C every hour until noon. Then it fell by 3 °C every hour.**

**(a)** The temperature at noon was ☐ °C

**(b)** The temperature at 3 pm was ☐ °C

**(c)** At what time before noon was it 29 °C? ☐ am

Working: _____

## Think harder

Hints are upside down. Try first!

**1** **Sales grow by the same number of toys each week. Week 3: 52 toys. Week 8: 87 toys.**

**(a)** It sold ☐ toys in Week 1.

**(b)** It first sold more than 100 toys in Week ☐.

Working: _____

_____

**Hint.** From Week 3 to Week 8 is 5 weeks. How much is that each week?

**2** **A stall sold 200 plates in 4 days. On Day 2 it sold twice as many as on Day 1. On Day 3 it sold 15 more than on Day 2. On Day 4 it sold 25. Find the number of plates sold on Day 1.** ☐

Draw a bar model here.

_____

_____

**Hint.** Start with a bar for Day 1. Compare the other days with it.

**3** **Taxi A charges $4 to start and then $2 for every km. Taxi B charges $10 to start and then $1 for every km.**

**(a)** For ☐ km the 2 taxis cost the same.

**(b)** Then each fare is $ ☐

Working: _____

_____

_____

**Hint.** Make a table of both fares for 1 km, 2 km, 3 km.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Two graphs show the same shop's sales in January and February. Graph X starts at 0 with 20 for each line. Graph Y starts at 60 with 5 for each line. Graph X shows January 4 lines up and February 5 lines up.**

**(a)** The sales rose by ☐

**(b)** On graph Y, January is ☐ lines up and February is ☐ lines up.

**(c)** Which graph makes the rise look greater? X Y

Working: _____

**Hint.** Find the sales in January and February first.

**2** **A stall sold 340 plates in 5 days, Monday to Friday. Each day's sales beat the day before by 8 plates.**

**(a)** On Monday it sold ☐ plates.

**(b)** On Friday it sold ☐ plates.

Working: _____

**Hint.** Take away the extras: 0, 8, 16, 24 and 32.

★ Beyond Level 4: Average

**3** **The average is the total divided by the count of numbers.**

**(a)** The average of 3, 6 and 9 is ☐

**(b)** A stall's average over 4 days is 45 plates. Three days were 40, 50 and 38 plates. The fourth day was ☐ plates.

Working: _____

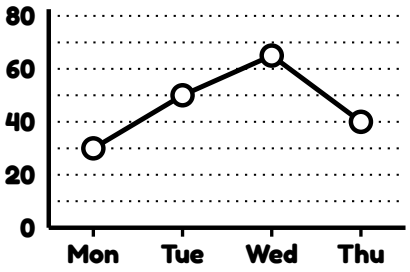
**Hint.** Total = average × the number of days.

## ✔ Check yourself

No hints. Try each one on your own.

**1**

**The graph shows the cakes a shop sold. Each line is worth 10. How many cakes were sold on Wednesday?** ☐




**2** **A shop sold 28 pens on Monday and 35 on Tuesday. The 3-day total was 99. Wednesday:** ☐ **pens**

Working: _____

**3** **Sales rise by the same amount each day. Day 1: 14. Day 4: 32. Day 6:** ☐

Working: _____

**4** **Both graphs show sales of 60 in May and 70 in June. A starts at 0 and B starts at 50. Which looks steeper?** **A** **B**

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# Learn it: Mixed puzzle heuristics

**My goal.** I can choose a way in to a puzzle: guess and check, suppose all are the same, work backwards, draw it or use a logic grid.

A puzzle needs a way in. Guess and check with a table: use each try to decide whether to go up or down. Suppose all are the same: then see how many swaps you need to fix the total.

## Worked example 1

**A shop has stools (3 legs) and chairs (4 legs). There are 11 of them with 38 legs in all. How many chairs are there?**

| | |
|---|---|
| 11 stools | 11 × 3 = 33 |
| Legs short | 38 – 33 = 5 |
| 1 swap adds | 1 leg |

1. Each swap from a stool to a chair adds 1 leg.
2. 5 legs short needs 5 swaps.

**Answer: 5 chairs and 6 stools. Check: 5 × 4 + 6 × 3 = 38.**

## Worked example 2

**Two next-door numbers multiply to 380. What are they?**

| Try | Product | Check |
|---|---|---|
| 18 × 19 | 342 | too small |
| 20 × 21 | 420 | too big |
| 19 × 20 | 380 | right |

1. Try a pair. Is the product too big or too small?
2. Use each try to choose the next one.

**Answer: The numbers are 19 and 20.**

## Ways in

1. Guess and check: keep a table of tries.
2. Suppose all are the same. Count swaps.
3. Work backwards: undo the last step first.
4. Check your answer against every fact.

## Good to know

- Draw a queue or a bar model.
- Check the answer against every clue.
- A swap changes the total by a fixed amount.
- Logic grid: ✓ for yes, × for no. One ✓ in each row and column.

## Your turn

Answer at the back of the book.

Try **A box has bees (6 legs) and spiders (8 legs). There are 10 animals with 66 legs.**

**How many spiders are there?** ☐

Name _____

## Start here

Do these first. Take your time.

**1** **Two next-door numbers multiply to make 506. What are they? Use the table for your tries.**

| First number | Next number | Product | Too big or small? |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |

The numbers are ☐ and ☐

**2** **Mei buys some $3 pens and some $5 notebooks. She buys at least 1 of each. She pays exactly $41.**

**(a)** How many different ways can she do it? ☐

**(b)** The way with the fewest pens: ☐ pens and ☐ notebooks

Notebooks, money left, pens: _____

My answer: _____

**3** **A farm has hens (2 legs) and cows (4 legs). There are 17 animals with 50 legs.**

**(a)** Suppose all 17 are hens. They have ☐ legs.

**(b)** That is ☐ legs short.

**(c)** Each swap to a cow adds 2 legs. There are ☐ cows.

Check: _____

## Start here

Keep going. Check your work.

**4** **A shed has bicycles (2 wheels) and tricycles (3 wheels). There are 13 of them with 33 wheels.**

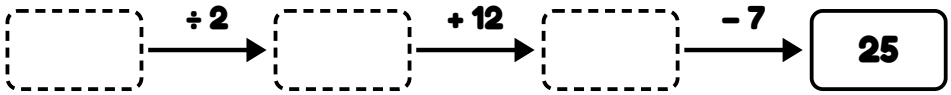
**(a)** The number of tricycles is ☐

**(b)** The number of bicycles is ☐

Working: _____

**5** **Mei had some stickers. She gave half of them to Ali. Then she bought 12 more. Then she lost 7. Now she has 25 stickers. Find how many she had at the start.**

☐




**6** **Kumar thinks of a number. He doubles it, adds 14 and then divides by 4. His answer is 10. What is his number?** ☐

Working: _____

_____

**7** **Ali, Ben, Mei and Siti each have a different pet: a cat, a dog, a fish or a rabbit. Ali's pet barks. Ben's pet has no fur. Mei's pet does not hop. Use the grid. Write ✓ for yes and × for no.**

| | cat | dog | fish | rabbit |
|---|---|---|---|---|
| Ali | | | | |
| Ben | | | | |
| Mei | | | | |
| Siti | | | | |

**(a)** Ali: ☐ **(b)** Ben: ☐ **(c)** Mei: ☐

**(d)** Siti: ☐

## Start here

Nearly there. You can do it.

**8** **Aisha stands in a queue. There are 8 people in front of her and 4 behind her.**

**(a)** The queue has ☐ people.

**(b)** Ben is in another queue. He has 4 people behind him and he is 6th in line.
That queue has ☐ people.

Draw the queue here.

**9** **Kumar is 4 times as old as his sister. Their ages add up to 45. How old is his sister?** ☐

Draw a bar model here.

**10** **Ravi has 9 coins. They are 20¢ coins and 50¢ coins. They are worth $3.00 altogether. Find the number of 50¢ coins.** ☐

Working: _____

**11** **Three next-door numbers add up to 87. What are they?** ☐, ☐, ☐

Working: _____

**12** **Who am I? My 2 digits add up to 12. Swapping them gives a number 18 greater than me.** ☐

Try: _____

## Think harder

Hints are upside down. Try first!

**1** **A shop sells pens in boxes of 2 and boxes of 5. Ali buys 12 boxes with 45 pens in all.**

**(a)** The number of boxes of 5 is ☐

**(b)** The number of boxes of 2 is ☐

Working: _____

_____

**Hint.** Suppose all 12 boxes hold 2 pens. How many pens are missing?

**2** **Ali, Ben, Kumar, Mei and Siti run a race. Nobody ties. Ali finishes before Mei. Mei finishes before Ben. Kumar finishes just after Siti. Ben does not finish last.**

**(a)** Places 1 to 5, in order: ☐ ☐ ☐ ☐ ☐

Try: _____

_____

**Hint.** Write what you know in 3 groups. Which group can go last?

**3** **Ravi saves $2 on Day 1, $4 on Day 2, $6 on Day 3 and so on. He saves $2 more each day.**

**(a)** His total first goes above $100 on Day ☐

**(b)** The total on that day is $ ☐

Day, saved, total: _____

_____

_____

**Hint.** Make a table: day, saved that day, total so far.

# ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **How many whole numbers from 1 to 200 have digits that add up to 5?** ☐

List: _____

**Hint.** Be systematic. How many 1-digit numbers fit?

**2** **Ali has 3 times as many 20¢ coins as 50¢ coins. His coins are worth $4.40 altogether. How many coins does he have in all?** ☐

Working: _____

_____

**Hint.** Try a small number of 50¢ coins first.

**3** **In a line of children, Ben is 11th from the left. Mei is 8th from the right. There are 5 children between Ben and Mei. How many children could be in the line? Find both answers.**

☐ **or** ☐

Draw the line here.

_____

**Hint.** Mei may be left or right of Ben.

★ **Beyond Level 4: Number cycles**

**4** **The ones digits of 3, 3 × 3, 3 × 3 × 3 and so on go round in a cycle.**

**(a)** **The ones digit of 3 × 3 × 3 × 3 × 3 (five 3s) is** ☐

**(b)** **The ones digit of 20 threes multiplied together is** ☐

Ones digits: _____

**Hint.** Write the ones digits of the first 6 products. Look for the cycle.

## ✔ Check yourself

No hints. Try each one on your own.

**1** **A garage has cars (4 wheels) and bikes (2 wheels). There are 10 of them with 28 wheels. How many cars are there?** ☐

Working: _____

**2** **Ali thinks of a number. He adds 8, doubles it and takes away 6. His answer is 30. What is his number?** ☐

Working: _____

**3** **There are 9 people in front of Siti in a line and 6 behind her. The line has** ☐ **people.**

Working: _____

**4** **Two next-door numbers multiply to make 272. What are they?** ☐ **and** ☐

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

Tick each box when you finish. Write your Check yourself score.

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| 1 | Factor Hunters | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 2 | Big Number Puzzles | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 3 | Operation Detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 4 | Fraction Frontier | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 5 | Decimal Detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 6 | Angle Turns | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 7 | Area Architects II | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 8 | Symmetry and Nets | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 9 | Data Detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 10 | Puzzle Day | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

**You finished the Math Inquirers Extension Workbook, Level 4.**

Date

Signed by a grown-up

# Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

# Unit 1 · Factor Hunters

**Your turn**
page 6

**Answer: 1, 3, 5, 9, 15, 45**
**Why:** 1 × 45, 3 × 15 and 5 × 9 are the pairs. 2 and 4 do not go into 45. After 5 comes 9, which is already used, so stop.

**START HERE**

**1**
page 7

**Answer: (a) 2 × 28, 4 × 14, 7 × 8 (b) 8 factors**
**Why:** 3, 5 and 6 do not go into 56. The pairs 1 × 56, 2 × 28, 4 × 14 and 7 × 8 give 8 factors. The next number, 8, is already used.

**2**
page 7

**Answer: 1, 2, 3, 6, 11, 22, 33, 66**
**Why:** 1 × 66, 2 × 33, 3 × 22 and 6 × 11. Then 7, 8, 9 and 10 do not go into 66, and 11 is already used.

**3**
page 7

**Answer: 2, 4, 8, 16**
**Why:** Team sizes are factors of 32 other than 1 and 32: 2, 4, 8 and 16. For example, 4 children in a team gives 8 teams.

**4**
page 7

**Answer: (a) True (b) False (c) True (d) True**
**Why:** (a) 7 × 13 = 91. (b) The multiples of 8 are 48, 56, so 54 is not one. (c) 9 × 12 = 108. (d) 13 × 5 = 65.

**5**
page 7

**Answer: (a) 9, 18, 27, 36, 45, 54 (b) 12, 24, 36, 48, 60, 72 (c) 36**
**Why:** Count in 9s and in 12s. The first number in both lists is 36 (4 nines and 3 twelves).

**6**
page 8

**Answer: (a) 1, 2, 4, 7, 14, 28 (b) 1, 2, 3, 6, 7, 14, 21, 42 (c) 1, 2, 7, 14 (d) 14**
**Why:** 28 = 1 × 28, 2 × 14, 4 × 7. 42 = 1 × 42, 2 × 21, 3 × 14, 6 × 7. The numbers in both lists are 1, 2, 7 and 14. The greatest is 14.

**7**
page 8

**Answer: (a) 15 bags (b) 2 red and 3 blue**
**Why:** The number of bags is a common factor of 30 and 45. The common factors are 1, 3, 5 and 15, so the greatest is 15. Each bag has 30 ÷ 15 = 2 red and 45 ÷ 15 = 3 blue.

**8**
page 8

**Answer: (a) 30 (b) 3**
**Why:** The least common multiple of 10 and 15 is 30 (10, 20, 30 and 15, 30). They jog together after 30, 60 and 90 days. The next, 120 days, is after the 100 days. That is 3 times.

**9**
page 8

**Answer: 84**
**Why:** A number in both the 6 and 7 times tables: 42, 84. Only 84 is between 50 and 100.

**10**
page 9

**Answer: (a) 8: 4, 14: 4, 18: 6, 27: 4, 32: 6, 49: 3 (b) 49**
**Why:** 8: 1, 2, 4, 8. 14: 1, 2, 7, 14. 18: 1, 2, 3, 6, 9, 18. 27: 1, 3, 9, 27. 32: 1, 2, 4, 8, 16, 32. 49: 1, 7, 49. Only 49 has 3 factors.

**11** page 9
**Answer: (a) 24 (b) 4 (c) 3**
**Why:** The least common multiple of 6 and 8 is 24 (6, 12, 18, 24 and 8, 16, 24). 24 ÷ 6 = 4 packs of pens and 24 ÷ 8 = 3 packs of rubbers.

**12** page 9
**Answer: 15, 20**
**Why:** The factors of 60 are 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. Of these, 15 and 20 are multiples of 5 between 10 and 30.

**13** page 9
**Answer: (a) 7 (b) 5 (c) 49**
**Why:** 64: 1, 2, 4, 8, 16, 32, 64 is 7 factors. 81: 1, 3, 9, 27, 81 is 5 factors. 49 = 7 × 7 pairs with itself, so it has 3 factors (odd). 72 has 12 factors.

## THINK HARDER

**1** page 10
**Answer: 72**
**Why:** A multiple of 3 and 8 is a multiple of 24: 24, 48, 72, 96. The digits add to 9 only for 72 (7 + 2). 7 is greater than 2.

**2** page 10
**Answer: 63**
**Why:** Without the 3 left over, the number is in the 4, 5 and 6 times tables, so it is a multiple of 60. 60 + 3 = 63 is between 50 and 100. The next one, 123, is too big.

**3** page 10
**Answer: (a) 36 teeth (b) A 3 times and B 2 times**
**Why:** The least common multiple of 12 and 18 is 36. Gear A: 36 ÷ 12 = 3 turns. Gear B: 36 ÷ 18 = 2 turns.

## GO FURTHER

**1** page 11
**Answer: 48**
**Why:** 48 has 1, 2, 3, 4, 6, 8, 12, 16, 24, 48: 10 factors. Numbers below 48 have fewer: for example 36 has 9, 40 has 8 and 42 has 8.

**2** page 11
**Answer: (a) 72 (b) 8 times**
**Why:** The least common multiple of 6, 8 and 9 is 72. 10 minutes is 600 seconds. 72 × 8 = 576 is less than 600 and 72 × 9 = 648 is more, so they flash together 8 times.

**3** page 11
**Answer: 17**
**Why:** List the amounts you can make: 4, 7, 8, 11, 12, 14, 15, 16, 18, 19, 20, 21, 22, 23 and every larger number. 17 is not on the list, and 17 − 7 = 10 and 17 − 14 = 3 are not multiples of 4. Every number from 18 up can be made.

**4** page 11
**Answer: Length 12 cm, width 7 cm**
**Why:** Half the perimeter is 19, so length + width = 19. The factor pairs of 84 are 1 and 84, 2 and 42, 3 and 28, 4 and 21, 6 and 14, 7 and 12. Only 7 and 12 add to 19.

## CHECK YOURSELF

**1** page 12
**Answer: 1, 2, 4, 13, 26, 52**
**Why:** 1 × 52, 2 × 26 and 4 × 13. Then 5 to 12 do not go into 52, and 13 is already used.

**2** page 12
**Answer: 18**
**Why:** Multiples of 6: 6, 12, 18. Multiples of 9: 9, 18. The first number in both lists is 18.

**3** page 12
**Answer: 8**
**Why:** The factors of 40 and 64 that are in both lists are 1, 2, 4, 8. The greatest is 8 (5 red and 8 blue in each set).

**4** page 12
**Answer: 14**
**Why:** The factors of 84 that are multiples of 7 are 7, 14, 21, 28, 42 and 84. Only 14 has digits that add up to 5 (1 + 4).

## Unit 2 · Big Number Puzzles

**Your turn** page 13
**Answer: 52 700**
**Why:** 52 718 is between 52 700 and 52 800. The halfway mark is 52 750. 52 718 is before it, so it rounds down to 52 700.

**START HERE**

**1** page 14
**Answer: 38 750, 38 700, 39 000, 70 050, 70 000, 70 000, 10 000, 10 000, 10 000**
**Why:** 38 746: the 6 ones round up to 38 750. To the nearest 100, 46 is under 50, so 38 700. To the nearest 1000, 746 is over 500, so 39 000. 70 049: 70 050, 70 000, 70 000. 9996 is 4 away from 10 000, so all three roundings are 10 000.

**2** page 14
**Answer: (a) 6000 (b) 60 000 (c) 60**
**Why:** 46 281 has 6 in the thousands place, so it is worth 6000. In 60 912 it is in the ten thousands place: 60 000. In 83 062 it is in the tens place: 60.

**3** page 14
**Answer: 42 007, 60 900, 8030**
**Why:** forty-two thousand and seven is 42 007: 42 thousands, no hundreds, no tens and 7 ones. sixty thousand, nine hundred is 60 900. eight thousand and thirty is 8030.

**4** page 14
**Answer: 58 430, 58 340, 58 304, 58 034**
**Why:** All four have 58 thousands. Compare the hundreds: 4, 3, 3, 0. The two with 3 hundreds are 58 340 and 58 304, and 340 is greater than 304.

**5** page 15
**Answer: (a) 24 300 (b) 24 800 (c) 25 000**
**Why:** There are 5 steps of 100 between 24 000 and 24 500. A is 3 steps after 24 000: 24 300. B is 3 steps after 24 500: 24 800. 24 800 is past the halfway mark 24 500, so it is closer to 25 000.

**6** page 15
**Answer: (a) 36 000 (b) 36 500**
**Why:** 36 482 is between 36 000 and 37 000, and it is before the halfway mark 36 500, so it rounds to 36 000. To the nearest 100 it is between 36 400 and 36 500, and 82 is more than 50, so 36 500.

**7** page 15
**Answer: (a) 11 500 (b) 12 499**
**Why:** 11 500 is halfway between 11 000 and 12 000 and rounds up to 12 000. 12 500 would round up to 13 000, so the largest is 12 499.

**8** page 15
**Answer: (a) 4650 (b) 4749 (c) 100**
**Why:** 4650 is halfway between 4600 and 4700, so it rounds up. 4750 would round to 4800, so the largest is 4749. From 4650 to 4749 there are 100 numbers.

**9** page 16
**Answer: (a) 20 000 (b) 20 090 (c) more**
**Why:** 4018 rounds to 4000, and 5 × 4000 = 20 000. The exact total is 5 × 4018 = 20 090. 4018 was rounded down, so the exact total is more.

**10** page 16
**Answer: 51 752**
**Why:** Numbers that round to 52 000 are from 51 500 to 52 499. With 7 hundreds the number must be 51 7__, because 52 7__ is too big. Then the tens and ones give 51 752.

**11** page 16
**Answer: (a) 97 420 (b) 20 479 (c) 94 207**
**Why:** Greatest: put the greatest digits first, 97 420. Least: 0 cannot go first, so start with 2: 20 479. Odd: the ones digit must be 7 or 9. Ending in 7 gives 94 207 and ending in 9 gives 74 209, so 94 207 is greater.

**12** page 16
**Answer: (a) 34 600, 35 600 (b) 65 900, 64 400 (c) 5009, 5012**
**Why:** (a) Add 1000 each time. (b) Take away 1500 each time: 67 400 – 1500 = 65 900, then 64 400. (c) Add 3 each time: 5006 + 3 = 5009, then 5012.

**13** page 16
**Answer: (a) < (b) < (c) > (d) >**
**Why:** (a) 45 009 has 0 tens, 45 090 has 9 tens, so it is smaller. (b) 8999 has 4 digits. (c) 62 100 has 1 hundred, 62 010 has none. (d) 30 000 + 700 is 30 700, which is greater than 30 070.

## THINK HARDER

**1** page 17
**Answer: 38 641**
**Why:** Just below 40 000: the greatest number that starts with 3 is 38 641, which is 1359 away. Just above 40 000: the least number that starts with 4 is 41 368, which is 1368 away. 38 641 is closer.

**2** page 17
**Answer: (a) 68 000 (b) 69 998**
**Why:** The smallest crowds are 40 500 and 27 500: 68 000. The largest are 41 499 and 28 499: 69 998.

**3** page 17
**Answer: (a) 74 986 (b) 65 012**
**Why:** Greatest: 74 999 is the top, but its digits repeat. Using 7, 4, then the greatest digits left gives 74 986. Least: 65 000 repeats the 0, so use 65 012.

### GO FURTHER

**1** page 18

**Answer: (a) 37 000 (b) 36 000 (c) 1445**
**Why:** 36 449 becomes 36 450, 36 500, then 37 000. In one go it is 36 000, because 449 is below 500. (c) 1445 becomes 1450, 1500, 2000, but in one go it is 1000. Any 4-digit number below 1445 gives the same answer both ways.

**2** page 18

**Answer: 271**
**Why:** Count the numbers with no 7. Each of 3 digits (000 to 999) can be any of 9 digits: 9 × 9 × 9 = 729, which includes 000 and leaves out 1000, which has no 7. So 729 numbers from 1 to 1000 have no 7, and 1000 – 729 = 271 have a 7.

**3** page 18

**Answer: (a) 8000 (b) 15**
**Why:** The numbers in places 1, 3, 5 ... are 12 000, 11 000, 10 000, 9000, 8000. The 9th is the 5th of these: 8000. The numbers in places 2, 4, 6 ... are 3, 6, 9, 12, 15. The 10th is the 5th of these: 15.

**4** page 18

**Answer: (a) No (b) 100**
**Why:** To the nearest 1000, the numbers are 6500 to 7499. To the nearest 100, they are 6950 to 7049. For example, 7400 is in the first list but rounds to 7400 to the nearest 100. The numbers in both lists are 6950 to 7049: 100 numbers.

### CHECK YOURSELF

**1** page 19

**Answer: 25 960, 26 000, 26 000**
**Why:** 25 962: the 2 ones round down to 25 960. The last two digits, 62, are over 50, so 26 000. The last three digits, 962, are over 500, so 26 000.

**2** page 19

**Answer: Least 8250, greatest 8349**
**Why:** 8250 is halfway between 8200 and 8300 and rounds up to 8300. 8350 would round up to 8400, so the greatest is 8349.

**3** page 19

**Answer: 10 487**
**Why:** The ones digit must be 7 or 1. Ending in 7 lets 1 go first: 10 487. Ending in 1 needs 4 first: 40 781. 10 487 is less.

**4** page 19

**Answer: 48 700, 47 200**
**Why:** Each number is 1500 less than the one before: 50 200 – 1500 = 48 700, then 47 200.

## Unit 3 · Operation Detectives

**Your turn** page 20

**Answer: 19**
**Why:** Rulers are 1 unit and pens are 5 units: 6 units in all. 6 units = 114, so 1 unit = 114 ÷ 6 = 19 rulers.

### START HERE

**1** page 21

**Answer: (a) 14 824 (b) 3995 (c) 1254 (d) 1630**
**Why:** (a) 3706 × 4 = 12 000 + 2800 + 24 = 14 824. (b) 85 × 47 = 85 × 40 + 85 × 7 = 3400 + 595 = 3995. (c) 6 × 1254 = 7524. (d) 5 × 1630 = 8150.

**2** page 21
**Answer: (a) 20 000 (b) 3000 (c) 4000**
**Why:** (a) 4986 rounds to 5000 and 5000 × 4 = 20 000. (b) 9120 rounds to 9000 and 9000 ÷ 3 = 3000. (c) 7020 rounds to 7000 and 2980 rounds to 3000, and 7000 − 3000 = 4000.

**3** page 21
**Answer: 963**
**Why:** 28 × 36 = 1008 chairs at first. 1008 − 45 = 963 chairs are left.

**4** page 21
**Answer: (a) 683 (b) 4**
**Why:** 5468 ÷ 8 = 683 with a remainder of 4, because 8 × 683 = 5464 and 5468 − 5464 = 4.

**5** page 22
**Answer: 77**
**Why:** Ducks are 1 unit and hens are 4 units: 5 units in all. 5 units = 385, so 1 unit = 385 ÷ 5 = 77 ducks.

**6** page 22
**Answer: (a) 19 (b) 28**
**Why:** Take away the extra $9: $47 − $9 = $38. Two equal parts are $38, so the ruler costs $38 ÷ 2 = $19. The pencil case costs $19 + $9 = $28.

**7** page 22
**Answer: 17**
**Why:** Work backwards. 31 × 3 = 93. 93 + 9 = 102. 102 ÷ 6 = 17.

**8** page 22
**Answer: (a) 4 (b) 4 (c) 4**
**Why:** 6 × 3 = 18, 18 + 12 = 30, 30 ÷ 3 = 10, 10 − 6 = 4. 15 × 3 = 45, 45 + 12 = 57, 57 ÷ 3 = 19, 19 − 15 = 4. Multiplying by 3 and then dividing by 3 gives back the number. The 12 ÷ 3 adds 4, and taking away the number leaves 4.

**9** page 23
**Answer: (a) 607 (b) 1116 (c) 2262 (d) 4495**
**Why:** (a) 4249 ÷ 7 = 607. (b) 124 × 9 = 1116. (c) 5000 − 2738 = 2262. (d) 6400 − 1905 = 4495.

**10** page 23
**Answer: 24**
**Why:** 6 × $3 = $18 and 4 × $2 = $8. The cost is $18 + $8 = $26. The change is $50 − $26 = $24.

**11** page 23
**Answer: (a) 13, 35 (b) 5, 17**
**Why:** (a) 4 + 9 = 13. Then 9 + 13 = 22 and 13 + 22 = 35. (b) 11 + 6 = 17 and 6 + 11 = 17. The first number is 11 − 6 = 5.

**12** page 23
**Answer: 240**
**Why:** Aisha has 1 unit and Kumar has 3 units: 4 units in all. 1 unit = 480 ÷ 4 = 120. Kumar has 360 and Aisha has 120. The difference is 2 units: 240.

## THINK HARDER

**1** page 24
**Answer: (a) 28 (b) 94**
**Why:** (a) 4 × 9 = 36 and 12 ÷ 3 = 4. Then 60 − 36 + 4 = 24 + 4 = 28. (b) Brackets first: 13 + 7 = 20. Then 5 × 20 = 100 and 100 − 6 = 94.

**2** page 24
**Answer: 7**
**Why:** The 4th number is 66 − 41 = 25. The 3rd is 41 − 25 = 16. The 2nd is 25 − 16 = 9. The 1st is 16 − 9 = 7. The chain is 7, 9, 16, 25, 41, 66.

**3** page 24
**Answer: (a) 18 (b) 90**
**Why:** Mei is 5 units and Lily is 1 unit. After the gift they are equal, so Mei had 4 units more than Lily, and 4 units is 36 + 36 = 72. 1 unit = 18. Lily had 18 and Mei had 90.

### GO FURTHER

**1** page 25
**Answer: (a) 9 × 4 − 6 (b) (7 + 3) × 8 − 4**
**Why:** (a) 9 × 4 = 36 and 36 − 6 = 30. The other signs do not give 30. (b) (7 + 3) × 8 − 4 = 10 × 8 − 4 = 76. Brackets round 3 × 8 give 7 + 24 − 4 = 27, and brackets round 8 − 4 give 7 + 3 × 4 = 19.

**2** page 25
**Answer: (a) 3 (b) 39**
**Why:** Plan A has $15, $23, $31, $39. Plan B has $0, $13, $26, $39. Both have $39 in week 3. Each week Plan B gains $5 on Plan A, and the start gap is $15, so 15 ÷ 5 = 3 weeks.

**3** page 25
**Answer: (a) 3 (b) 3**
**Why:** 4 × 7 = 28, which ends in 8, as needed. 3□4 × 7 = 2128 + 70 × □ must be 2□38. Trying 3 gives 2128 + 210 = 2338. Check: 334 × 7 = 2338. So both missing digits are 3.

### CHECK YOURSELF

**1** page 26
**Answer: 23 640 and 1305**
**Why:** 4728 × 5 = 23 640 (4728 × 10 = 47 280, then half). 9135 ÷ 7 = 1305, because 7 × 1305 = 9135.

**2** page 26
**Answer: 116**
**Why:** 15 × 12 = 180, so he saves $180. $180 − $64 = $116.

**3** page 26
**Answer: 220**
**Why:** Blue is 1 unit and red is 4 units: 5 units. 1 unit = 275 ÷ 5 = 55. Red is 4 × 55 = 220.

**4** page 26
**Answer: 21**
**Why:** Work backwards. 94 + 20 = 114. 114 ÷ 3 = 38. 38 − 17 = 21.

## Unit 4 · Fraction Frontier

**Your turn** page 27
**Answer: 19/5**
**Why:** 3 wholes are 3 × 5 = 15 fifths. 15 fifths and 4 more fifths make 19 fifths: 19/5.

### START HERE

**1** page 28
**Answer: (a) 17 (b) 2 5/6**
**Why:** There are 6 + 6 + 5 = 17 sixths shaded. 2 whole bars are shaded and 5 of the 6 parts of the third bar: 2 5/6.

**2** page 28
**Answer: (a) 4 3/4 (b) 3 5/6 (c) 3 7/8 (d) 8**
**Why:** (a) 19 ÷ 4 = 4 remainder 3. (b) 23 ÷ 6 = 3 remainder 5. (c) 31 ÷ 8 = 3 remainder 7. (d) 40 ÷ 5 = 8 with nothing left, so it is exactly 8.

**3** page 28
**Answer: (a) 17/5 (b) 16/3 (c) 23/8**
**Why:** (a) 3 × 5 + 2 = 17 fifths. (b) 5 × 3 + 1 = 16 thirds. (c) 2 × 8 + 7 = 23 eighths.

**4** page 28
**Answer: (a) 20 (b) 16**
**Why:** 9 units = 36, so 1 unit = 36 ÷ 9 = 4. Red is 5 units: 5 × 4 = 20 pencils. Not red is 4 units: 4 × 4 = 16 pencils.

**5** page 29
**Answer: (a) 2/3 (b) 1 1/3 (c) 8/3**
**Why:** Each step is 1/3. A is 2 steps from 0: 2/3. B is 1 whole and 1 more step: 1 1/3. C is 8 steps from 0, so 8/3, which is 2 2/3.

**6** page 29
**Answer: (a) 84 (b) 48**
**Why:** 3 units = 36 pages, so 1 unit = 12 pages. The whole book is 7 units: 7 × 12 = 84 pages. Left is 4 units: 4 × 12 = 48 pages.

**7** page 29
**Answer: (a) > (b) > (c) < (d) >**
**Why:** (a) 3/4 = 6/8, and 6/8 is more than 5/8. (b) 4/6 = 8/12, and 8/12 is more than 7/12. (c) 3/8 is less than 1/2 and 5/9 is more than 1/2. (d) Both have 5 parts, and sixths are larger than eighths.

**8** page 29
**Answer: 5/12, 1/2, 2/3, 3/4**
**Why:** Write each as twelfths: 3/4 = 9/12, 2/3 = 8/12, 5/12, 1/2 = 6/12. In order: 5/12, 6/12, 8/12, 9/12.

**9** page 30
**Answer: (a) 8 (b) 4 (c) 3 (d) 15**
**Why:** (a) 3 × 4 = 12, so 2 × 4 = 8. (b) 15 ÷ 5 = 3, so 20 ÷ 5 = 4. (c) 12 ÷ 3 = 4, so 9 ÷ 3 = 3. (d) 4 × 3 = 12, so 5 × 3 = 15.

**10** page 30
**Answer: (a) 3 3/8 pizzas (b) 40 (c) 5**
**Why:** (a) 27 ÷ 8 = 3 remainder 3, so 3 3/8 pizzas. (b) 5 × 8 = 40. (c) 4 pizzas are 32 slices, and 32 − 27 = 5.

**11** page 30
**Answer: (a) 8 (b) 1**
**Why:** Each metre has 3 thirds. 2 2/3 m is 8 thirds, so there are 8 pieces. After 5 pieces, 3 pieces are left, and 3 thirds make 1 m.

**12** page 30
**Answer: 24**
**Why:** 1/4 of 48 is 12, so she has 48 − 12 = 36 left. 1/3 of 36 is 12. Then 36 − 12 = 24 stickers.

**13** page 30
**Answer: (a) 20 (b) 18 (c) 5**
**Why:** (a) 5 × 4 = 20. (b) 3 × 6 = 18. (c) 2 wholes make 4 halves, and 1/2 makes 1 more half: 5 halves.

## THINK HARDER

**1** page 31
**Answer: 26/7**
**Why:** Between 3 and 4 with denominator 7: 22/7, 23/7, 24/7, 25/7, 26/7, 27/7. Even numerators: 22, 24, 26. 3 1/2 is 24.5/7, so closer to 4 means 25/7 or more. Only 26/7 fits.

**2** page 31
**Answer: 90**
**Why:** 2/5 = 6/15 and 1/3 = 5/15, so she spent 11/15. 4/15 is left. 4 units = $24, so 1 unit = $6 and 15 units = $90.

**3** page 31
**Answer: 120**
**Why:** After Monday, 2/3 of the book is left. On Tuesday she reads 3/4 of that, so 1/4 of it is left. 1/4 of 2/3 is 1/6 of the book. 1/6 = 20 pages, so the book has 6 × 20 = 120 pages.

### GO FURTHER

**1** page 32
**Answer: (a) 6 (b) 10**
**Why:** 1/4 = 5/20 and 3/5 = 12/20. The fractions in between are 6/20, 7/20, 8/20, 9/20, 10/20 and 11/20: 6 fractions. 10/20 = 1/2.

**2** page 32
**Answer: (a) 1/2 + 1/3 (b) 1/2 + 1/4 (c) 1/2 + 1/5**
**Why:** (a) 5/6 − 1/2 = 5/6 − 3/6 = 2/6 = 1/3. (b) 3/4 − 1/2 = 1/4. (c) 7/10 − 1/2 = 7/10 − 5/10 = 2/10 = 1/5.

**3** page 32
**Answer: 45**
**Why:** 6 girls are 1/3 of the girls, so there are 18 girls. 18 girls are 2/5 of the class, so 1/5 is 9 children and the class has 5 × 9 = 45 children.

**4** page 32
**Answer: (a) < (b) > (c) <**
**Why:** (a) 7/9 is 2/9 short of 1 and 8/10 is 2/10 short of 1. 2/9 is more than 2/10, so 7/9 is less. (b) 1/12 is less than 1/10, so 11/12 is greater. (c) 1/15 is more than 1/20, so 14/15 is less.

### CHECK YOURSELF

**1** page 33
**Answer: 4 5/6**
**Why:** 29 ÷ 6 = 4 remainder 5. That is 4 wholes and 5 sixths: 4 5/6.

**2** page 33
**Answer: 43/8**
**Why:** 5 × 8 + 3 = 43, so 5 3/8 = 43/8.

**3** page 33
**Answer: 5/8**
**Why:** In twenty-fourths, 5/8 = 15/24 and 7/12 = 14/24. 15/24 is greater, so 5/8 is greater.

**4** page 33
**Answer: 63**
**Why:** 4 units = 28, so 1 unit = 7. The bag is 9 units: 9 × 7 = 63 beads.

## Unit 5 · Decimal Detectives

**Your turn** page 34
**Answer: 8.5**
**Why:** 8.46 is between 8.4 and 8.5. The halfway mark is 8.45. 8.46 is after it, so it rounds up to 8.5.

## START HERE

**1** page 35

**Answer: (a) 5.3 (b) 5.8**
**Why:** From 5 to 6 there are 10 steps, so each step is 0.1. A is 3 steps after 5: 5.3. B is 8 steps after 5: 5.8.

**2** page 35

**Answer: (a) 0.43 (b) 0.48 (c) D**
**Why:** From 0.4 to 0.5 there are 10 steps, so each step is 0.01. C is 3 steps after 0.4: 0.43. D is 8 steps after 0.4: 0.48. 0.48 is past the halfway mark 0.45, so it rounds to 0.5.

**3** page 35

**Answer: (a) 3.427**
**Why:** From 3.42 to 3.43 there are 10 steps, so each step is 0.001. E is 7 steps after 3.42: 3.427.

**4** page 36

**Answer: (a) < (b) > (c) = (d) <**
**Why:** (a) 4.06 has 0 tenths and 4.6 has 6 tenths. (b) 0.38 has 8 hundredths and 0.308 has 0 hundredths. (c) 12.50 is 12.5 with a 0 added. (d) 7.09 is 7.09 and 7.1 is 7.10, and 10 hundredths is more than 9.

**5** page 36

**Answer: 5.025, 5.052, 5.205, 5.25, 5.5**
**Why:** With 3 decimal places: 5.205, 5.250, 5.052, 5.500, 5.025. Compare the tenths: 0 (5.025, 5.052), then 2 (5.205, 5.250), then 5 (5.500). Then compare the hundredths.

**6** page 36

**Answer: (a) 0.06 (b) 6 (c) 0.006**
**Why:** In 4.562 the 6 is in the hundredths place: 0.06. In 6.043 it is in the ones place: 6. In 0.936 it is in the thousandths place: 0.006.

**7** page 36

**Answer: (a) 7 (b) 6.8 (c) 6.85**
**Why:** (a) 6.847 is between 6 and 7 and is past 6.5, so 7. (b) 6.847 is between 6.8 and 6.9. The halfway mark is 6.85, so it rounds to 6.8. (c) 6.847 is between 6.84 and 6.85. The halfway mark is 6.845, so it rounds up to 6.85.

**8** page 36

**Answer: 0.75**
**Why:** $1.85 + $2.40 = $4.25. $5.00 − $4.25 = $0.75, which is 75¢.

**9** page 37

**Answer: (a) 5.10 (b) 7.50**
**Why:** Take away the extra $2.40: $12.60 − $2.40 = $10.20. Two equal parts are $10.20, so Kumar has $10.20 ÷ 2 = $5.10. Ravi has $5.10 + $2.40 = $7.50.

**10** page 37

**Answer: (a) 3.6 (b) 0.9 (c) 3.2**
**Why:** 4.1 − 0.5 = 3.6 kg for the 4 packs. 3.6 ÷ 4 = 0.9 kg for 1 pack. After 1 pack is taken out, the mass is 4.1 − 0.9 = 3.2 kg.

**11** page 37

**Answer: (a) 7.05 (b) 5.56 (c) 10 (d) 1.2**
**Why:** (a) 4.70 + 2.35 = 7.05. (b) 9.20 − 3.64 = 5.56. (c) 2.5 × 4 = 10. (d) 7.2 ÷ 6 = 1.2, because 6 × 1.2 = 7.2.

**12** page 37

**Answer: (a) 0.7 (b) 0.19 (c) 3.5 (d) 2.75**
**Why:** (a) 7 tenths is 0.7. (b) 19 hundredths is 0.19. (c) 1/2 = 5/10 = 0.5, so 3.5. (d) 3/4 = 75/100 = 0.75, so 2.75.

## THINK HARDER

**1** page 38

**Answer: (a) 4.265 (b) 4.274**
**Why:** 4.265 is halfway between 4.26 and 4.27, so it rounds up to 4.27. 4.275 would round up to 4.28, so the greatest is 4.274.

**2** page 38

**Answer: 19.64**
**Why:** Just below 20 start with 19: 19.64 is 0.36 below 20 and 19.46 is 0.54 below. Numbers starting 14 or 16 are further away, and numbers above 20 (41.69, 46.19, 49.16 and so on) are much further.

**3** page 38

**Answer: (a) 2.00 (b) 4.40**
**Why:** If every mug cost the same as a cup, the 5 items would cost $17.20 – 3 × $2.40 = $17.20 – $7.20 = $10.00. So a cup costs $10.00 ÷ 5 = $2.00 and a mug costs $4.40.

## GO FURTHER

**1** page 39

**Answer: 0.24**
**Why:** The juice that was drunk has a mass of 1.26 – 0.58 = 0.68 kg, and that is 2 thirds of the juice. 1 third is 0.34 kg, so all the juice is 1.02 kg. The empty bottle is 1.26 – 1.02 = 0.24 kg.

**2** page 39

**Answer: 6**
**Why:** If all 12 coins were 10¢, the total would be $1.20. The real total is $2.40 more. Changing a 10¢ coin into a 50¢ coin adds 40¢. $2.40 ÷ 40¢ = 6 coins. So she has 6 coins of 50¢ and 6 coins of 10¢.

**3** page 39

**Answer: (a) 4.20 (b) 6.00**
**Why:** The extras are $0 + $0.60 + $1.20 + $1.80 = $3.60. The rest, $20.40 – $3.60 = $16.80, is shared equally: $4.20 each. The amounts are $4.20, $4.80, $5.40 and $6.00.

**4** page 39

**Answer: (a) 1.3 (b) 35**
**Why:** The pattern adds 0.05 each time. The 20th number is 0.35 + 19 × 0.05 = 0.35 + 0.95 = 1.30. For 2.05, the number of steps from 0.35 is (2.05 – 0.35) ÷ 0.05 = 34, so it is the 35th number.

## CHECK YOURSELF

**1** page 40

**Answer: 6.007, 6.07, 6.17, 6.7**
**Why:** With 3 decimal places: 6.070, 6.700, 6.007, 6.170. In order: 6.007, 6.070, 6.170, 6.700.

**2** page 40

**Answer: 8, 7.8, 7.85**
**Why:** 7.846 is past 7.5, so 8. It is below 7.85, so 7.8. It is past 7.845, so 7.85.

**3** page 40

**Answer: 4.60**
**Why:** 4 × $1.35 = $5.40. $10.00 – $5.40 = $4.60.

**4** page 40
**Answer: Least 3.55, greatest 3.64**
**Why:** 3.55 is halfway between 3.5 and 3.6, so it rounds up to 3.6. 3.65 would round up to 3.7, so the greatest is 3.64.

## Unit 6 · Angle Turns

**Your turn** page 41
**Answer: S**
**Why:** 135° is 3 steps of 45°. Clockwise from NE: E, SE, S. He faces S.

**START HERE**

**1** page 42
**Answer: (a) 25 (b) 65 (c) 130**
**Why:** Each angle starts at 0 on the horizontal arm. a = 25°, b = 65° and c = 130°. Answers within 2° are fine. Angle c is greater than a right angle, so a good estimate is more than 90°.

**2** page 42
**Answer: (a) 52 (b) smaller**
**Why:** One arm lies along the baseline at 0 on the left. Read the scale that starts at 0 on the left arm: 52°. A right angle is 90°, and 52° is smaller. The other scale says 128°, which is the angle measured from the other arm.

**3** page 42
**Answer: (a) 60 (b) 75 (c) 100 (d) 6**
**Why:** The angles add. ∠AOC = 25° + 35° = 60°. ∠BOD = 35° + 40° = 75°. ∠AOD = 25° + 35° + 40° = 100°. The angles are AOB, BOC, COD, AOC, BOD and AOD: 6 angles.

**4** page 43
**Answer: (a) N (b) NW (c) NW (d) NW**
**Why:** (a) 90° is 2 steps. From W clockwise: NW, N. (b) 180° is 4 steps. From SE: S, SW, W, NW. (c) 270° is 6 steps. From NE clockwise: E, SE, S, SW, W, NW. (d) Go back the other way: 135° is 3 steps clockwise from S: SW, W, NW.

**5** page 43
**Answer: (a) 150 (b) 120 (c) 30**
**Why:** Each space between numbers is 30°. At 5 o'clock the hands are 5 spaces apart: 150°. At 8 o'clock they are 4 spaces apart the short way: 120°. At 11 o'clock they are 1 space apart: 30°.

**6** page 43
**Answer: (a) 135 (b) clockwise**
**Why:** Clockwise from W: NW, N, NE is 3 steps, 135°. Anticlockwise from W: SW, S, SE, E, NE is 5 steps, 225°. The smallest turn is 135° clockwise.

**7** page 44
**Answer: (a) 53 (b) 43**
**Why:** (a) a + 37° = 90°, so a = 90° − 37° = 53°. (b) The three angles add to 90°: b = 90° − 19° − 28° = 43°.

**8** page 44
**Answer: (a) E (b) 270**
**Why:** 135° anticlockwise from E: NE, N, NW. 90° clockwise from NW: N, NE. 45° clockwise from NE: E. It faces E. The turns add to 135° + 90° + 45° = 270°.

**9** page 44
**Answer: (a) 15 (b) 75**
**Why:** The corner is 90° and is made of 1 + 5 = 6 equal parts. 90° ÷ 6 = 15° for 1 part. The greater angle is 5 × 15° = 75°.

**10** page 44
**Answer: (a) 15 (b) 30 (c) 45 (d) 10**
**Why:** The minute hand turns 6° each minute. 90° ÷ 6 = 15. 180° ÷ 6 = 30. 270° ÷ 6 = 45. 60° ÷ 6 = 10.

## THINK HARDER

**1** page 45
**Answer: (a) 48 (b) 67**
**Why:** (a) 47° + 85° = 132°, and 180° − 132° = 48°. (b) 95° + 120° + 78° = 293°, and 360° − 293° = 67°.

**2** page 45
**Answer: (a) 45 (b) 135**
**Why:** At 7:30 the minute hand is on the 6 and the hour hand is halfway between 7 and 8, 1 and a half spaces away: 45°. At 1:30 the hour hand is 4 and a half spaces from the 6: 4 × 30° + 15° = 135°.

**3** page 45
**Answer: (a) SE (b) 495**
**Why:** In steps of 45°: +2, −3, +5, −1. From N: 2 steps is E, then 3 steps back is NW, then 5 steps forward is S, then 1 step back is SE. The turns add to 90° + 135° + 225° + 45° = 495°.

## GO FURTHER

**1** page 46
**Answer: (a) 15 (b) 3**
**Why:** Each small angle is 150° ÷ 5 = 30°. Angles made of 1 part: 5. Of 2 parts: 4. Of 3 parts: 3. Of 4 parts: 2. Of 5 parts: 1. In all 5 + 4 + 3 + 2 + 1 = 15. A right angle is 3 parts of 30°, and there are 3 of them.

**2** page 46
**Answer: (a) 8 (b) W**
**Why:** Each turn is 3 steps. The directions are SE, W, NE, S, NW, E, SW, N. It faces N again after 8 turns. 50 turns is 6 lots of 8, which is 48 turns, and 2 more turns. After 48 turns it faces N. 2 more turns take it to SE and then W.

**3** page 46
**Answer: (a) 10 (b) 60**
**Why:** The 3 angles are 1, 2 and 6 equal parts. That is 9 parts in all, and 9 parts = 90°, so 1 part = 10°. The smallest is 10°, the middle is 20° and the greatest is 60°.

**4** page 46
**Answer: (a) 4 (b) 4, 5, 7 and 8 o'clock**
**Why:** The angle at 1 to 12 o'clock is 30°, 60°, 90°, 120°, 150°, 180°, 150°, 120°, 90°, 60°, 30°, 0°. Those between 100° and 160° are 120°, 150°, 150° and 120°, at 4, 5, 7 and 8 o'clock.

## CHECK YOURSELF

**1** page 47
**Answer: 150**
**Why:** At 7 o'clock the hands are 5 spaces apart the short way. 5 × 30° = 150°.

**2** page 47
**Answer: NW**
**Why:** 135° is 3 steps of 45°. Clockwise from S: SW, W, NW.

**3** page 47
**Answer: 49**
**Why:** The corner is 90°. 90° − 41° = 49°.

**4** page 47
**Answer: 150**
**Why:** From 5:10 to 5:35 is 25 minutes. 25 × 6° = 150°.

## Unit 7 · Area Architects II

**Your turn** page 48
**Answer: 54**
**Why:** The rectangle has an area of 12 × 5 = 60 cm². The corner has an area of 3 × 2 = 6 cm². 60 − 6 = 54 cm².

**START HERE**

**1** page 49
**Answer: (a) a = 4 cm, b = 5 cm (b) 68 (c) 38**
**Why:** (a) a = 11 − 7 = 4 cm and b = 8 − 3 = 5 cm. (b) Split: 11 × 3 + 7 × 5 = 68 cm². Or 11 × 8 − 4 × 5 = 68 cm². (c) Push the steps out: 2 × (11 + 8) = 38 cm.

**2** page 49
**Answer: (a) 108 (b) 20 (c) 88**
**Why:** (a) 12 × 9 = 108 cm². (b) The corner is 5 cm by 4 cm: 5 × 4 = 20 cm². (c) 108 − 20 = 88 cm².

**3** page 49
**Answer: (a) 1 and 58, 2 and 32, 4 and 22 (b) 22**
**Why:** 28 ÷ 28 = 1, 28 ÷ 14 = 2 and 28 ÷ 7 = 4. The perimeters are 2 × (28 + 1) = 58, 2 × (14 + 2) = 32 and 2 × (7 + 4) = 22. The least is 22 cm, from the shape that is nearest to a square.

**4** page 50
**Answer: (a) 5 (b) 10, 12, 12 (c) A**
**Why:** Each shape has 5 tiles, so the area is 5 cm². Count the edges round each shape. A: 10 cm. B: 5 + 5 + 1 + 1 = 12 cm. C: the plus sign has 12 cm. A is the most compact shape, so it has the least perimeter.

**5** page 50
**Answer: (a) 40 (b) 60**
**Why:** (a) Push the steps out. The staircase fits in a rectangle that is 12 cm by 8 cm, so the perimeter is 2 × (12 + 8) = 40 cm. (b) The rows are 12, 9, 6 and 3 cm long and each is 2 cm high: 2 × (12 + 9 + 6 + 3) = 2 × 30 = 60 cm².

**6** page 50
**Answer: (a) 16 cm by 13 cm (b) 100**
**Why:** (a) The frame adds 2 cm on each side: 12 + 2 + 2 = 16 and 9 + 2 + 2 = 13. (b) The whole is 16 × 13 = 208 cm². The photo is 12 × 9 = 108 cm². The frame is 208 − 108 = 100 cm².

**7** page 51
**Answer: (a) 30, 30 (b) 26, 22**
**Why:** (a) 10 × 3 = 30 cm² and 5 × 6 = 30 cm². Moving pieces does not change the area. (b) Strip: 2 × (10 + 3) = 26 cm. Rectangle: 2 × (5 + 6) = 22 cm. The perimeter changed.

**8** page 51
**Answer: (a) 8 (b) 34**
**Why:** (a) 72 ÷ 9 = 8 cm. (b) 2 × (9 + 8) = 34 cm.

**9** page 51
**Answer: (a) 24 (b) 16**
**Why:** (a) 12 ÷ 2 = 6 tiles along the length and 8 ÷ 2 = 4 tiles along the width. 6 × 4 = 24 tiles. (b) The tiles not along the edge make a block of 4 by 2 = 8 tiles. 24 − 8 = 16 dark tiles.

## THINK HARDER

**1** page 52

**Answer: 100**
**Why:** Each rectangle is 1 whole side and 1 half side, twice round: 2 × (s + s/2) = 3 × s. So 3 sides of the square make 30 cm, and each side is 10 cm. The area of the square is 10 × 10 = 100 cm².

**2** page 52

**Answer: (a) 49 (b) 98**
**Why:** (a) length + width = 14, so the greatest area is a 7 m square: 49 m². (b) The fence is 2 widths and 1 length: 2w + l = 28. w = 7, l = 14 gives 98 m². w = 6 or w = 8 gives 96 m². The greatest is 98 m².

**3** page 52

**Answer: 70**
**Why:** Move the paths to the edges. The grass is then one rectangle that is 12 − 2 = 10 m long and 9 − 2 = 7 m wide: 10 × 7 = 70 m².

## GO FURTHER

**1** page 53

**Answer: 75**
**Why:** The edge of the L shape has 8 sides of squares. 40 ÷ 8 = 5 cm for 1 side. Each square has an area of 5 × 5 = 25 cm², so the L shape has 3 × 25 = 75 cm².

**2** page 53

**Answer: Length 6 cm, width 3 cm**
**Why:** Width 3, length 6: area 18 cm² and perimeter 2 × 9 = 18 cm. Width 1 or 2 never works (width 2 gives 2l and 2l + 4). Widths of 4 or more give only the 4 by 4 square. So 6 cm by 3 cm.

**3** page 53

**Answer: (a) 36 (b) 40**
**Why:** (a) 10 × 2 + 10 × 2 = 40, but the 2 cm by 2 cm square in the middle is counted twice: 40 − 4 = 36 cm². (b) Each of the 4 arms is 4 cm long and 2 cm wide. Round one arm the edge is 4 + 2 + 4 = 10 cm, so 4 arms make 40 cm.

**4** page 53

**Answer: (a) 28 (b) 16**
**Why:** (a) The light tiles make a 6 by 6 block: 36. 64 − 36 = 28 dark tiles. (b) A floor n tiles long has 4 × n − 4 dark tiles. 4 × n − 4 = 60 gives n = 16.

## CHECK YOURSELF

**1** page 54

**Answer: 43**
**Why:** Split into 9 × 3 = 27 and 4 × 4 = 16: 27 + 16 = 43 cm². Or 9 × 7 − 5 × 4 = 63 − 20 = 43 cm².

**2** page 54

**Answer: 34**
**Why:** The width is 60 ÷ 12 = 5 cm. The perimeter is 2 × (12 + 5) = 34 cm.

**3** page 54

**Answer: 38**
**Why:** The whole is 12 cm by 9 cm = 108 cm². The photo is 10 × 7 = 70 cm². 108 − 70 = 38 cm².

**4** page 54

**Answer: 20**
**Why:** The pairs are 1 × 24, 2 × 12, 3 × 8 and 4 × 6, with perimeters 50, 28, 22 and 20. The least is 20 cm, from 4 cm by 6 cm.

# Unit 8 · Symmetry and Nets

**Your turn**
page 55

**Answer:** 3
**Why:** 4 + 3 = 7, so 3 is opposite 4.

**START HERE**

**1**
page 56

**Answer:** A and C
**Why:** A: the row of 4 makes the walls, with a lid above and a base below. It folds. B: a 2 by 2 block puts 4 squares at one corner, but a cube has only 3 faces at a corner. D: 5 in a row wrap round and the 5th covers the 1st. C folds.

**2**
page 56

**Answer: (a)** R **(b)** T **(c)** U
**Why:** In the row of 3 squares P, Q, R, the 2 ends P and R are opposite. Q and T are opposite (1 square between them going round the step). The last pair is S and U.

**3**
page 56

**Answer:** 5, 4, 6 (row by row)
**Why:** Find the opposite squares. 2 and the third square of the row are opposite, so that square is 7 − 2 = 5. 3 and the fourth square are opposite, so it is 7 − 3 = 4. 1 and the bottom square are opposite, so it is 7 − 1 = 6.

**4**
page 57

**Answer:** 24
**Why:** There are 12 shaded squares on the left of the line. Each needs a partner on the right, so 12 more squares are shaded. 12 + 12 = 24.

**5**
page 57

**Answer: (a)** 1 **(b)** 2 **(c)** 4 **(d)** 0
**Why:** A has 1 line (down the middle). B has 2 (along its length and across). C, the plus sign, has 4 (up-down, across and 2 diagonals). D has none: no fold makes its halves match.

**6**
page 57

**Answer: (a)** 2
**Why:** Only the shaded square in the bottom row has no partner. For the up-down line its partner is in the bottom row on the right. For the across line its partner is in the top row. Each fix works for one line: 2 ways.

**7**
page 58

**Answer: (a)** 1 **(b)** 3 **(c)** 14
**Why:** Opposite faces add up to 7. The bottom is 7 − 6 = 1. The back is 7 − 4 = 3. The 4 side faces are 2 opposite pairs, and 7 + 7 = 14.

**8**
page 58

**Answer:** 54
**Why:** Each square has an area of 3 × 3 = 9 cm². The net has 6 squares: 6 × 9 = 54 cm².

**9**
page 58

**Answer:** 20
**Why:** There are 10 shaded squares above the line. Each needs a partner below the line, so 10 more squares are shaded. 10 + 10 = 20.

**10**
page 58

**Answer: (a)** 60 **(b)** 150
**Why:** (a) 12 × 5 = 60 cm. (b) Each face is 5 × 5 = 25 cm², and the net has 6 faces: 6 × 25 = 150 cm².

## THINK HARDER

**1** page 59

**Answer: (a) 2 (b) 3**
**Why:** (a) A face shares an edge with 4 faces. Two opposite red faces do not share an edge, but a third red face would touch one of them. So the most is 2. (b) 3 faces at one corner are not opposite each other, but a 4th face would be opposite one of them. So the most is 3.

**2** page 59

**Answer: (a) 1, 4, 5 (b) 2, 3, 6**
**Why:** The pairs are 1 and 6, 2 and 5, 3 and 4. She sees one number from each pair, so no two of the 3 numbers add up to 7. The only choice with a total of 10 is 1, 5 and 4. The hidden faces are opposite: 6, 2 and 3.

**3** page 59

**Answer: 10**
**Why:** The 3 shaded squares are 3 from the middle column, or 1 from the middle column and 1 pair. 3 in the middle column: 1 way. 1 in the middle column (3 choices) and 1 pair (3 rows to choose): $3 \times 3 = 9$ ways. $1 + 9 = 10$.

## GO FURTHER

**1** page 60

**Answer: (a) 8 (b) 12 (c) 6 (d) 1**
**Why:** 3 painted faces: the 8 corner cubes. 2 faces: 1 in the middle of each of the 12 edges. 1 face: 1 in the middle of each of the 6 faces. No paint: the 1 cube in the middle. $8 + 12 + 6 + 1 = 27$.

**2** page 60

**Answer: (a) 17 (b) 16**
**Why:** A single square has a line of symmetry when it is on the middle row, the middle column or a diagonal. Middle row and column: $5 + 5 - 1 = 9$ squares. The diagonals add 8 more: 17. The middle square has 4 lines, so the other 16 have exactly 1.

**3** page 60

**Answer: (a) 6 (b) 1**
**Why:** Start: top 1, bottom 6, north 2, south 5, east 3, west 4. Roll north: top 5, bottom 2, north 1, south 6. Roll east: top 4, bottom 3, east 5, west 2. Roll north: the north face 1 lands down, so the bottom is 1 and the top is the old south face, 6.

## CHECK YOURSELF

**1** page 61

**Answer: No**
**Why:** Both extra squares are above the row of 4, on the same side. When the row wraps round, they land on the same face. So the net does not fold.

**2** page 61

**Answer: 2**
**Why:** It can be folded along the middle in 2 ways, and the halves match. The diagonals do not work.

**3** page 61

**Answer: bottom 5, back 4**
**Why:** Opposite faces add up to 7. The bottom is $7 - 2 = 5$ and the back is $7 - 3 = 4$.

**4** page 61
**Answer:** 96
**Why:** Each face is 4 × 4 = 16 cm². The net has 6 faces: 6 × 16 = 96 cm².

## Unit 9 · Data Detectives

**Your turn** page 62
**Answer:** 80
**Why:** 2 lines are worth 2 × 5 = 10. 70 + 10 = 80 books.

**START HERE**

**1** page 63
**Answer: (a) 60 (b) 45 (c) 75**
**Why:** (a) The Thursday point is on the 60 line. (b) The Tuesday point is halfway between 40 and 50: 45. (c) 30 + 45 = 75.

**2** page 63
**Answer: (a) Friday to Saturday, up by 40 (b) Tuesday to Wednesday, down by 20**
**Why:** The changes are +15, −20, +35, −5 and +40. The greatest increase is Friday to Saturday: 95 − 55 = 40. The greatest decrease is Tuesday to Wednesday: 45 − 25 = 20.

**3** page 63
**Answer: (a) 47, 41, 74, 112, 117, 229 (b) 4B**
**Why:** 4A: 85 − 38 = 47. 4B: 70 − 29 = 41. 4C: 33 + 41 = 74. Week 1: 38 + 41 + 33 = 112. Week 2: 47 + 29 + 41 = 117. Total: 112 + 117 = 229, and 85 + 70 + 74 = 229 too. 4B borrowed 41 in Week 1 and only 29 in Week 2.

**4** page 64
**Answer: (a) 60 (b) 10 (c) 7 (d) 8**
**Why:** (a) The dashed Mei line is on 60 in Week 3. (b) Ali's line goes up by 10 each week. (c) The lines meet in Week 7 at $80. (d) In Week 8 Ali has $90 and Mei has $85, so Week 8 is the first week Ali has more.

**5** page 64
**Answer: (a) 5 (b) 110**
**Why:** Day 1 to Day 4 is 3 days, and 27 − 12 = 15, so she reads 15 ÷ 3 = 5 more each day. The days are 12, 17, 22, 27 and 32. The total is 12 + 17 + 22 + 27 + 32 = 110 pages.

**6** page 65
**Answer: (a) 48 (b) 10 (c) P**
**Why:** (a) P: 88 − 40 = 48. (b) Q: 52 − 42 = 10. (c) P went up by 48 and Q by 10. Bakery Q's line looks steeper only because its scale starts at 40 and each line is worth 2.

**7** page 65
**Answer: (a) 60 (b) 90 (c) No**
**Why:** (a) Monday is on the 60 line. (b) Tuesday is on the 90 line. (c) Double 60 is 120, and 90 is less. The point looks more than twice as high because the scale starts at 40, not at 0.

**8** page 65
**Answer: (a) 35 (b) 26 (c) 9**
**Why:** (a) From 7 am to noon is 5 hours: 25 + 5 × 2 = 35 °C. (b) From noon to 3 pm is 3 hours: 35 − 3 × 3 = 26 °C. (c) 29 − 25 = 4 °C more, and 4 ÷ 2 = 2 hours after 7 am, which is 9 am.

## THINK HARDER

**1** page 66

**Answer: (a) 38 (b) 10**
**Why:** 87 − 52 = 35 over 5 weeks, so sales go up by 7 each week. Week 1 is 2 weeks before Week 3: 52 − 14 = 38. Week 9 is 94 and Week 10 is 101, so Week 10 is the first week above 100.

**2** page 66

**Answer: 32**
**Why:** Take away Day 4 and the extra 15: 200 − 25 − 15 = 160. Day 1 is 1 unit, Day 2 is 2 units and Day 3 is 2 units: 5 units = 160. 1 unit = 32.

**3** page 66

**Answer: (a) 6 (b) 16**
**Why:** Taxi A: $6, $8, $10, $12, $14, $16 for 1 to 6 km. Taxi B: $11, $12, $13, $14, $15, $16. The fares are equal at 6 km: $4 + 6 × $2 = $16 and $10 + 6 × $1 = $16.

## GO FURTHER

**1** page 67

**Answer: (a) 20 (b) 4 lines and 8 lines (c) Y**
**Why:** On X, January is 4 × 20 = 80 and February is 5 × 20 = 100, so sales rose by 20. On Y: (80 − 60) ÷ 5 = 4 lines and (100 − 60) ÷ 5 = 8 lines. Graph Y's line goes up 4 lines and X's goes up 1 line, so Y makes the rise look greater.

**2** page 67

**Answer: (a) 52 (b) 84**
**Why:** The extras over Monday are 0 + 8 + 16 + 24 + 32 = 80. Without them, 5 equal days make 340 − 80 = 260, so Monday is 260 ÷ 5 = 52. Friday is 52 + 32 = 84. The days are 52, 60, 68, 76 and 84.

**3** page 67

**Answer: (a) 6 (b) 52**
**Why:** (a) 3 + 6 + 9 = 18, and 18 ÷ 3 = 6. (b) The total for 4 days is 4 × 45 = 180. The 3 known days make 40 + 50 + 38 = 128. The fourth day is 180 − 128 = 52 plates.

## CHECK YOURSELF

**1** page 68

**Answer: 65**
**Why:** The Wednesday point is halfway between the 60 line and the 70 line: 65 cakes.

**2** page 68

**Answer: 36**
**Why:** 28 + 35 = 63 pens on Monday and Tuesday. 99 − 63 = 36 pens on Wednesday.

**3** page 68

**Answer: 44**
**Why:** From Day 1 to Day 4 is 3 days, and 32 − 14 = 18, so sales go up by 6 each day. Day 6 is 32 + 2 × 6 = 44.

**4** page 68

**Answer: B**
**Why:** Graph B's scale starts at 50, so the same 10 more looks like a greater rise. Sales rose by 10 on both graphs.

# Unit 10 · Puzzle Day

**Your turn**
page 69

**Answer:** 3
**Why:** Suppose all 10 are bees: 60 legs. That is 6 short. Each swap to a spider adds 2 legs, so 3 swaps. There are 3 spiders and 7 bees.

**START HERE**

**1**
page 70

**Answer: 22 and 23**
**Why:** 20 × 21 = 420 is too small. 25 × 26 = 650 is too big. 22 × 23 = 506. Use each try to decide the next one: 21 × 22 = 462 is too small, so go up.

**2**
page 70

**Answer: (a) 3 (b) 2 pens and 7 notebooks**
**Why:** Try the number of notebooks. 1 leaves $36 = 12 pens. 4 leaves $21 = 7 pens. 7 leaves $6 = 2 pens. The other numbers (0, 2, 3, 5, 6, 8) leave amounts that are not multiples of $3. So there are 3 ways, and the fewest pens is 2 with 7 notebooks.

**3**
page 70

**Answer: (a) 34 (b) 16 (c) 8**
**Why:** (a) 17 × 2 = 34. (b) 50 − 34 = 16. (c) 16 ÷ 2 = 8 swaps, so 8 cows and 9 hens.

**4**
page 71

**Answer: (a) 7 (b) 6**
**Why:** Suppose all 13 are bicycles: 26 wheels. That is 7 short. Each swap to a tricycle adds 1 wheel, so 7 tricycles and 13 − 7 = 6 bicycles.

**5**
page 71

**Answer: 40**
**Why:** Work backwards. Undo the loss of 7: 25 + 7 = 32. Undo the 12 she bought: 32 − 12 = 20. Undo giving away half: 20 × 2 = 40.

**6**
page 71

**Answer: 13**
**Why:** Work backwards. 10 × 4 = 40. 40 − 14 = 26. 26 ÷ 2 = 13.

**7**
page 71

**Answer: (a) dog (b) fish (c) cat (d) rabbit**
**Why:** Ali's pet barks, so it is the dog. The fish is the only pet with no fur, so Ben has the fish. A rabbit hops, so Mei does not have the rabbit. The cat and the rabbit are left. Mei has the cat and Siti has the rabbit.

**8**
page 72

**Answer: (a) 13 (b) 10**
**Why:** (a) 8 people in front, Aisha herself and 4 behind: 8 + 1 + 4 = 13. (b) Ben is 6th, so 5 people are in front of him. 5 + 1 + 4 = 10.

**9**
page 72

**Answer:** 9
**Why:** The sister is 1 unit and Kumar is 4 units: 5 units = 45, so 1 unit = 9. The sister is 9 and Kumar is 36.

**10**
page 72

**Answer:** 4
**Why:** Suppose all 9 are 20¢: 180¢. That is 120¢ short. Each swap to a 50¢ coin adds 30¢. 120 ÷ 30 = 4 swaps. So 4 coins of 50¢ and 5 coins of 20¢.

**11** page 72
**Answer: 28, 29, 30**
**Why:** The middle number is the one in the middle of the 3, so 87 ÷ 3 = 29. The numbers are 28, 29 and 30.

**12** page 72
**Answer: 57**
**Why:** Try pairs of digits that add up to 12: 39, 48, 57, 66, 75, 84, 93. The swapped numbers are 93, 84, 75, 66, 57, 48, 39. The swapped number is 18 more only for 57 and 75: 75 – 57 = 18. The number must be the smaller one, 57.

## THINK HARDER

**1** page 73
**Answer: (a) 7 (b) 5**
**Why:** If all 12 boxes held 2 pens, there would be 24 pens. 45 – 24 = 21 pens are missing. Each box of 5 instead of 2 adds 3 pens, so 21 ÷ 3 = 7 boxes of 5. The other 12 – 7 = 5 boxes hold 2.

**2** page 73
**Answer: (a) Ali, Mei, Ben, Siti, Kumar**
**Why:** Ali, Mei and Ben finish in that order. Siti and Kumar finish side by side, Siti first. Ben is not last, so Siti and Kumar must come after Ben: Ali, Mei, Ben, Siti, Kumar.

**3** page 73
**Answer: (a) 10 (b) 110**
**Why:** The totals are $2, $6, $12, $20, $30, $42, $56, $72, $90 and $110 for Days 1 to 10. On Day 9 the total is $90, which is not above $100. On Day 10 it is $110.

## GO FURTHER

**1** page 74
**Answer: 11**
**Why:** 1-digit: 5. 2-digit: 14, 23, 32, 41, 50. 3-digit, starting with 1: the other 2 digits add to 4: 104, 113, 122, 131, 140. 200 has a digit sum of 2. In all 1 + 5 + 5 = 11.

**2** page 74
**Answer: 16**
**Why:** A group of 3 coins of 20¢ and 1 coin of 50¢ is worth 60¢ + 50¢ = 110¢. $4.40 = 440¢ and 440 ÷ 110 = 4 groups. Each group has 4 coins, so he has 16 coins: 12 coins of 20¢ and 4 coins of 50¢.

**3** page 74
**Answer: 12 or 24**
**Why:** Mei right of Ben: she is 6 places after him, 11 + 6 = 17th from the left, so there are 17 + 7 = 24 children. Mei left of Ben: she is 11 – 6 = 5th from the left, so there are 5 + 7 = 12 children. Both fit.

**4** page 74
**Answer: (a) 3 (b) 1**
**Why:** The products are 3, 9, 27, 81, 243, 729, 2187. The ones digits 3, 9, 7, 1 go round in a cycle of 4. (a) The 5th product ends in 3. (b) 20 is 5 whole cycles, so the 20th ends in 1.

## CHECK YOURSELF

**1** page 75

**Answer: 4**
**Why:** Suppose all 10 are bikes: 20 wheels, 8 short. Each swap to a car adds 2 wheels, so 4 cars and 6 bikes.

**2** page 75

**Answer: 10**
**Why:** Work backwards. 30 + 6 = 36. 36 ÷ 2 = 18. 18 – 8 = 10.

**3** page 75

**Answer: 16**
**Why:** 9 in front, Siti herself and 6 behind: 9 + 1 + 6 = 16.

**4** page 75

**Answer: 16 and 17**
**Why:** 15 × 16 = 240 is too small and 17 × 18 = 306 is too big. 16 × 17 = 272.